

Euclid: The Era of *One-Hundred Thousand* Strong Lenses

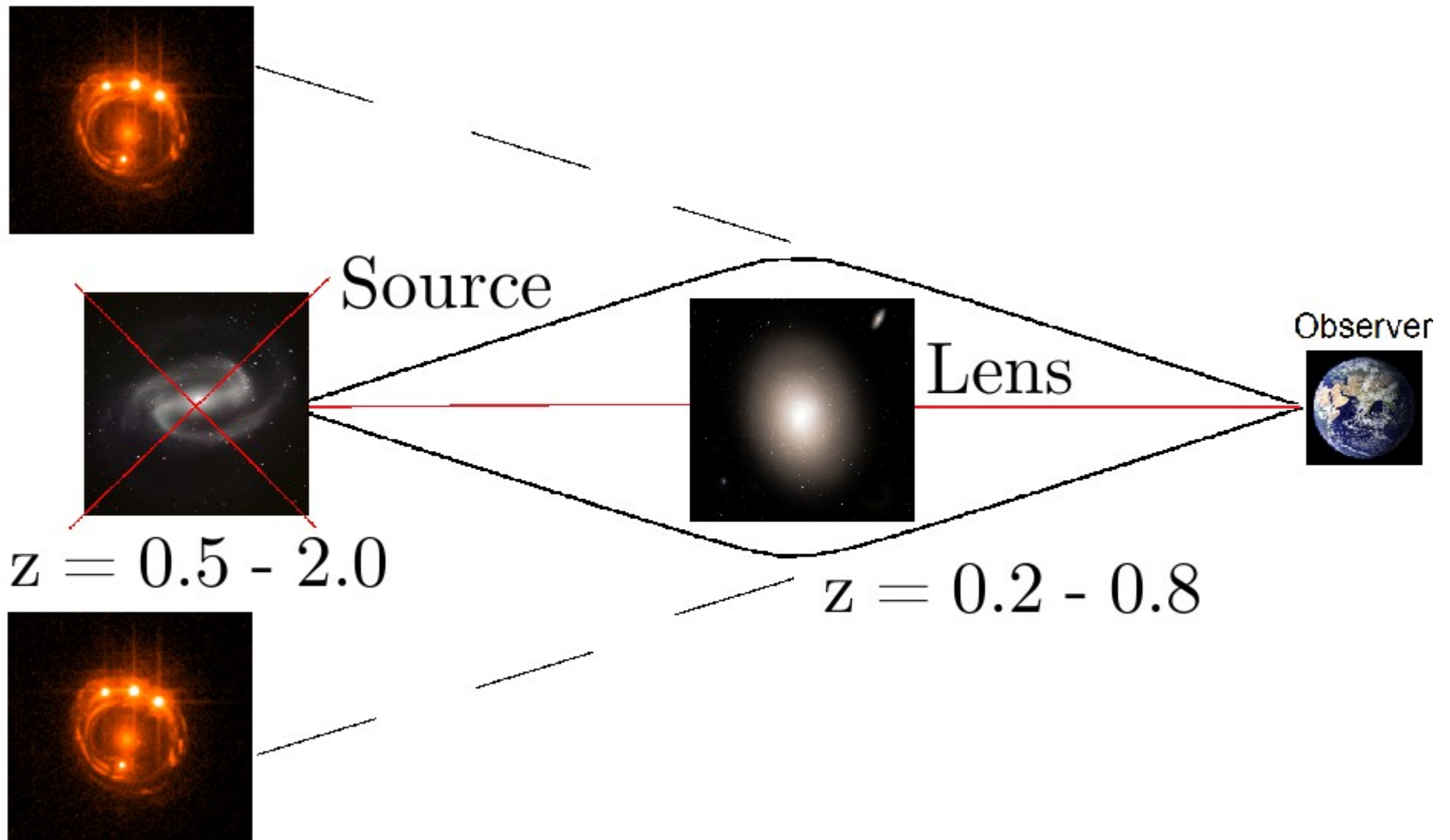
James Nightingale

Simon Dye

Overview

- **Galaxy-Scale Strong Lensing Overview.**
- **AutoLens:**
 - Source Reconstruction
 - Lens Light Modeling
 - Lens Mass Modeling
 - Automated Framework
- **Results - ER 0047-2808**
- **Euclid**

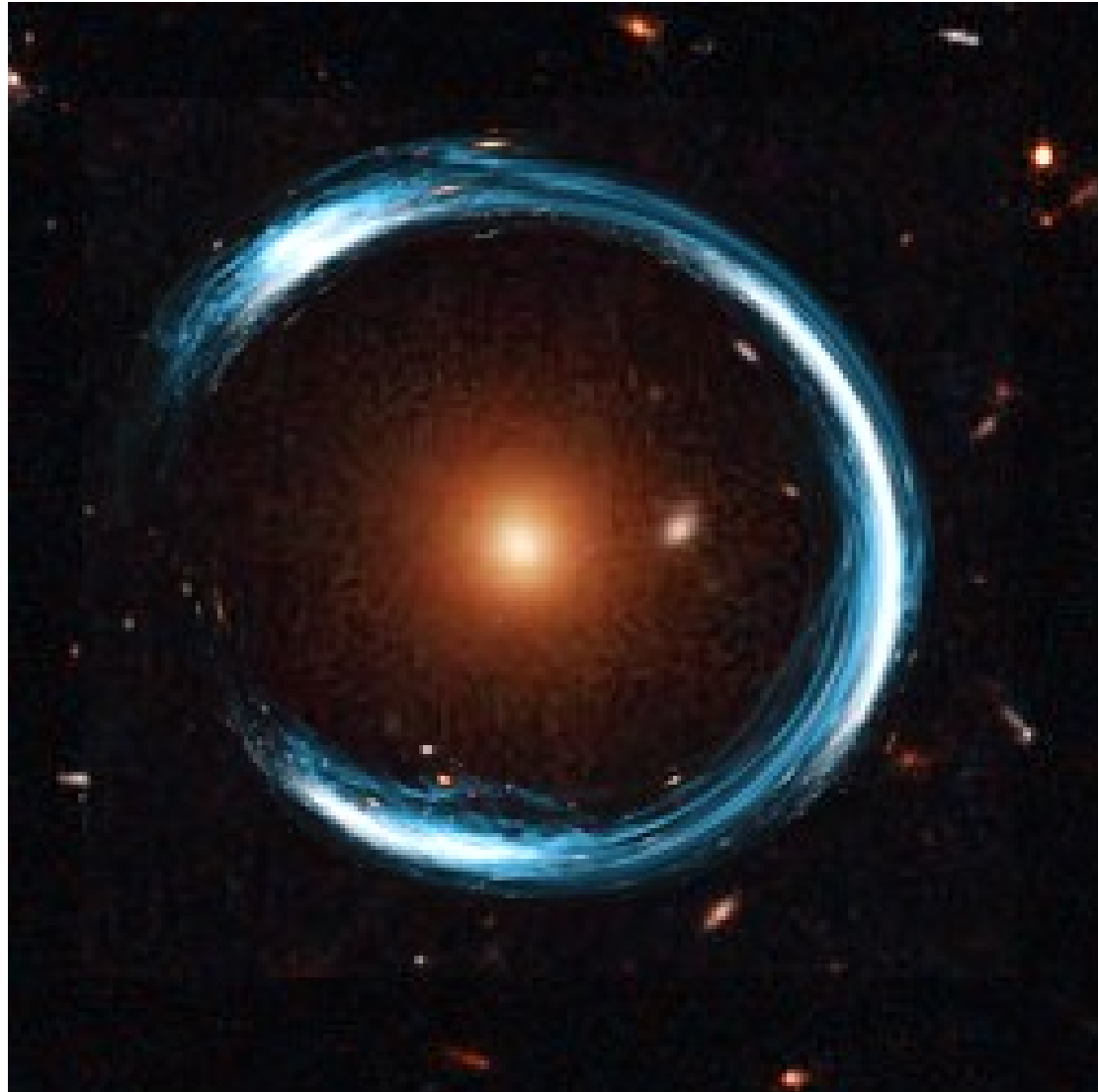
Strong Lensing



'Simple' Strong Lens Modeling

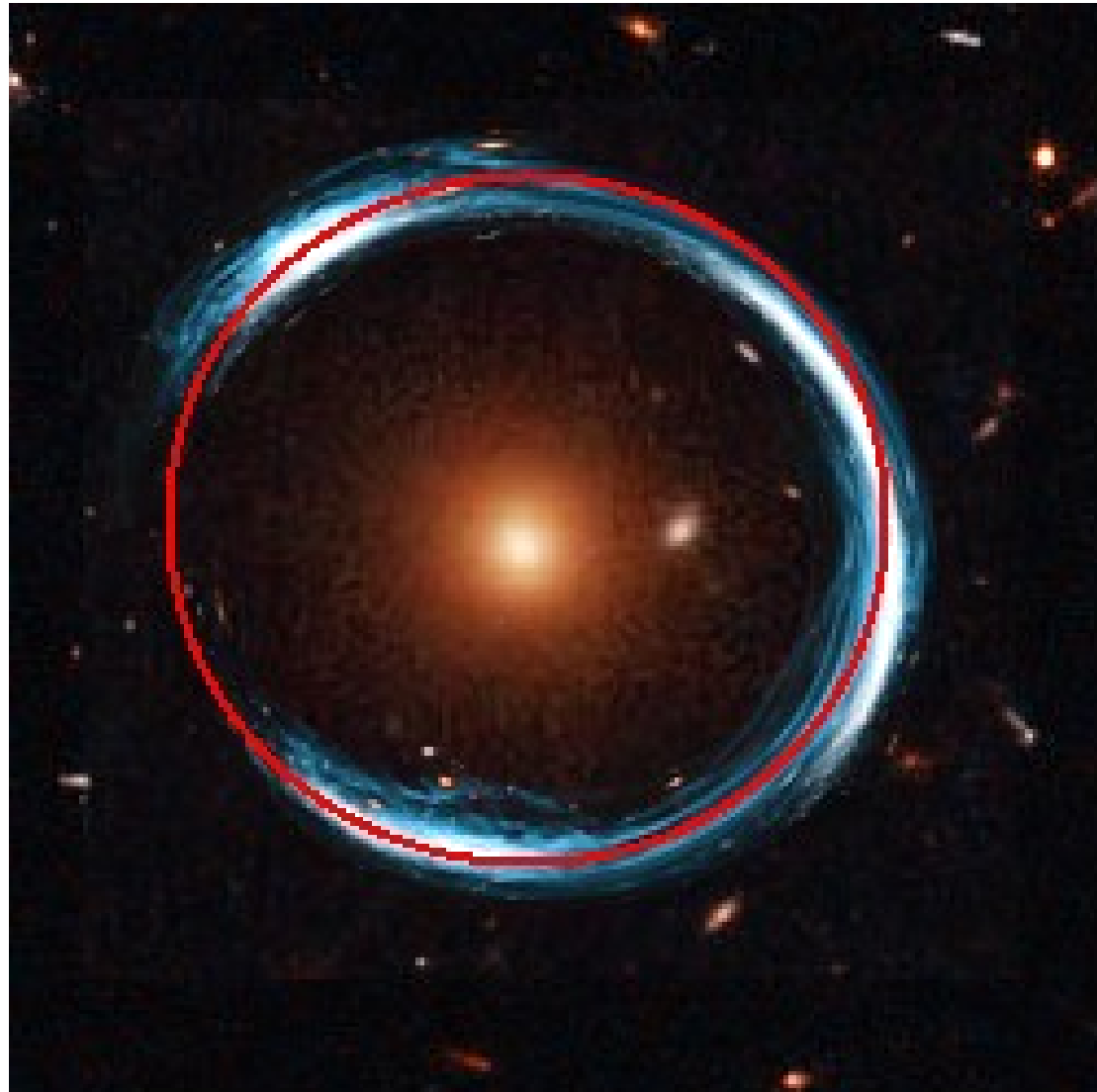
'Simple' Strong Lensing

- The 'simple' approach:
- **Uses** - the source position and source + lens redshifts.



'Simple' Strong Lensing

- **The 'simple' approach:**
- **Uses** - the source position and source + lens redshifts.
- **Measures** - The Einstein Mass, M_{ein} . Total mass within source aperture.
- Combine with your favorite astrophysical probe e.g. stellar dynamics, population synthesis.



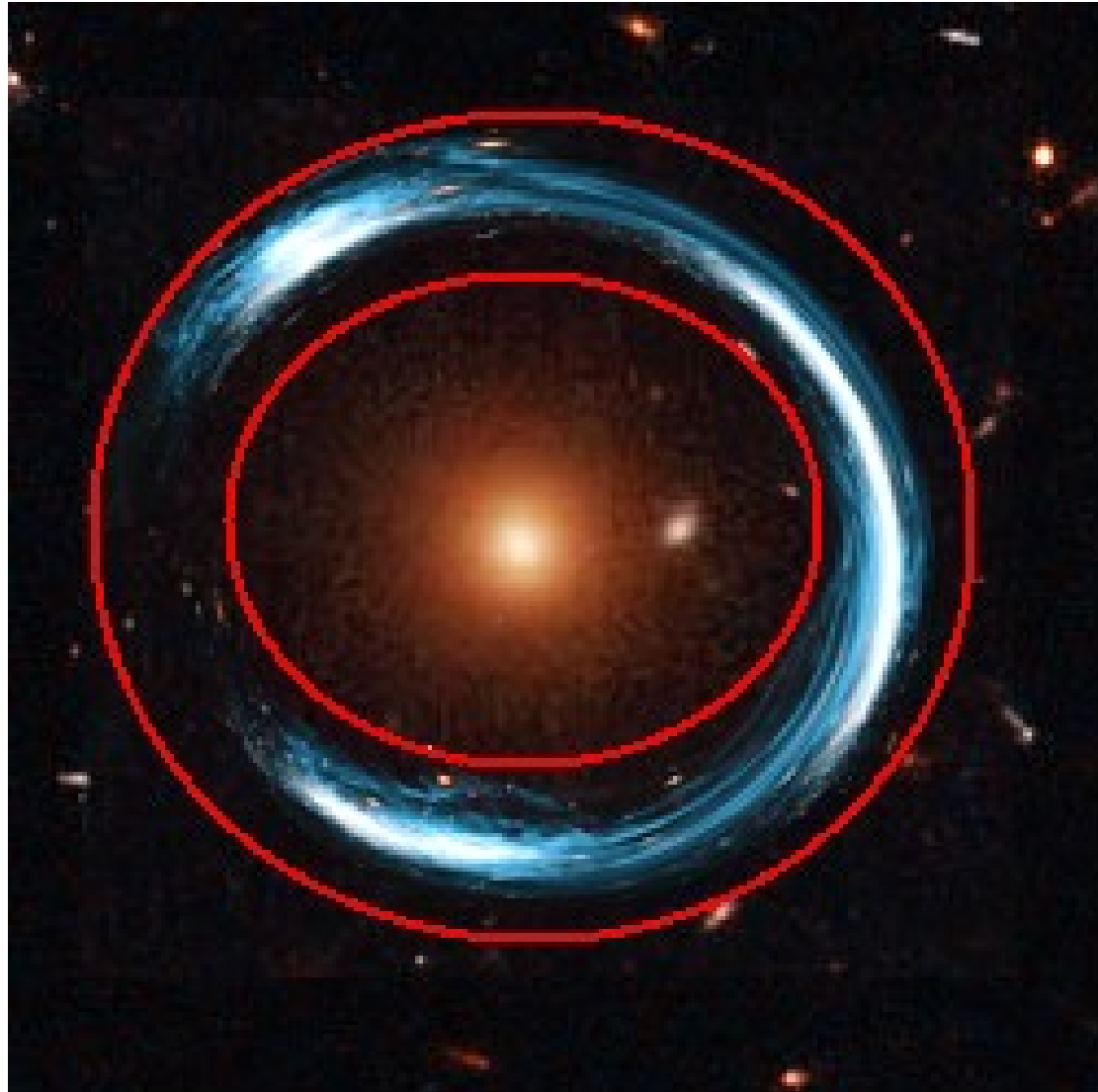
'Simple' Strong Lensing - Science

- **Combine with Stellar Dynamics:**
 - Two independent total mass measurements at two different radii.
 - Measure the lens galaxy's inner density slope
- **Combine with stellar population synthesis:**
 - Independent stellar mass measurement.
 - Tests the initial mass function.

'Advanced' Strong Lens Modeling

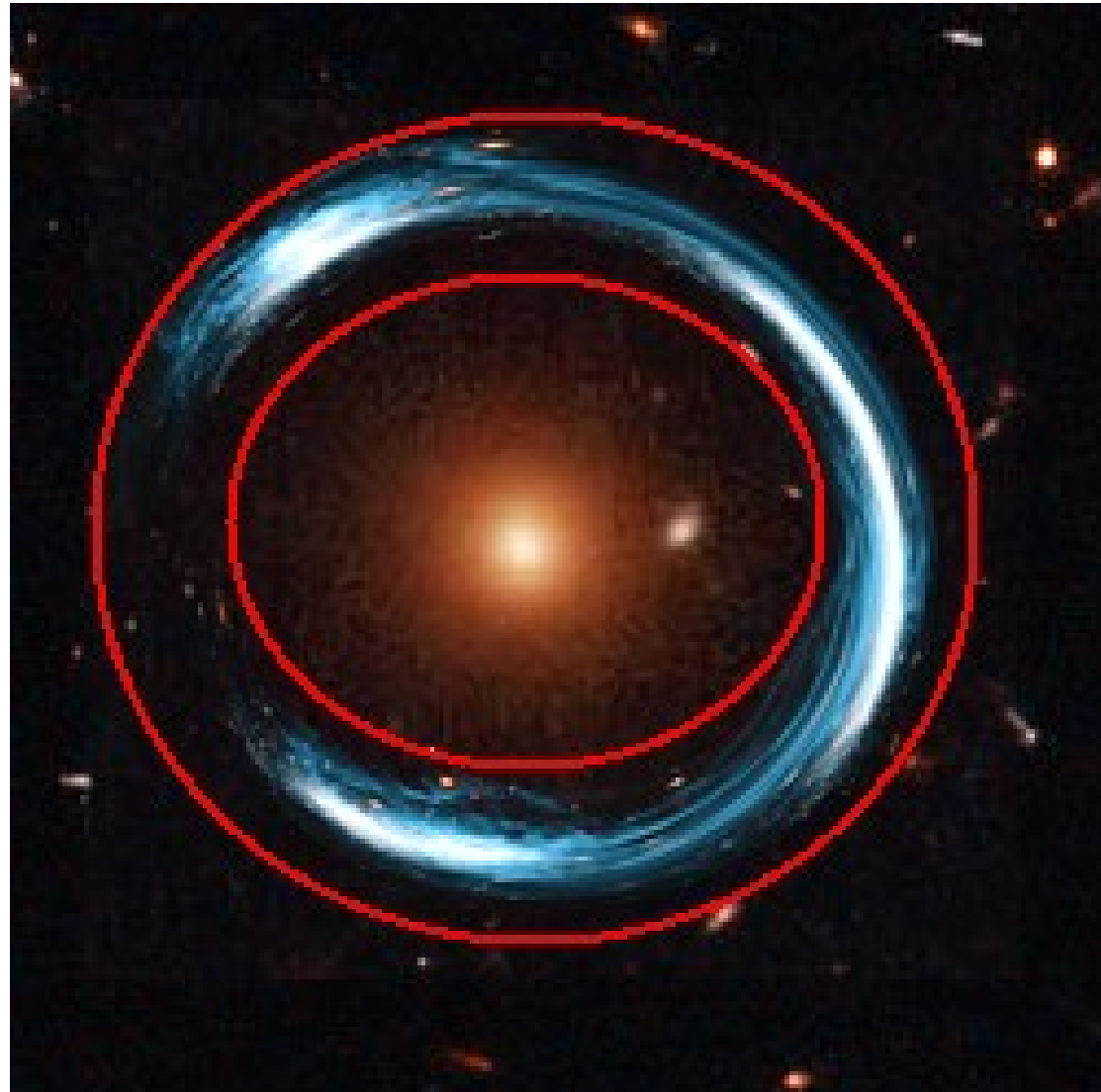
What does a strong lens tell us?

- **The 'Advanced' approach:**
- **Uses** - The extended source light distribution.



What does a strong lens tell us?

- **The 'Advanced' approach:**
- **Uses** - The extended source light distribution.
- **Measures (direct)** – The lens's mass distribution, at the Einstein Radius, R_{ein} .
- **Measures (indirect)** – The lens's mass distribution within R_{ein} .



'Advanced' Strong Lensing - Science

- **Cold Dark Matter Substructure Detection (Vegetti et al.) :**
 - Look for non-symmetric lensing features in lensed source.
 - Extragalactic test of Λ CDM substructure hierarchy.
- **Measure the Hubble Constant (Suyu et al.):**
 - Measure time-delays from strongly lensed quasars.
 - Combine with slope measurement from lens model.
- **Study highly-magnified sources (Lots of Authors):**
 - Need to reconstruct the source if you want to study it!

Modeling a Strong Lens

- **The 'Advanced' approach – Need to simultaneously model:**
 - The source's intrinsic light distribution.
 - The lens galaxy's mass distribution.
 - The lens galaxy's light profile (or subtract it beforehand).
- **And, ideally, you'd be able to do this on large samples of strong lenses.**

Modeling a Strong Lens

- **The 'Advanced' approach – Need to simultaneously model:**
 - The source's intrinsic light distribution.
 - The lens galaxy's mass distribution.
 - The lens galaxy's light profile (or subtract it beforehand).
- **And, ideally, you'd be able to do this on large samples of strong lenses.**
- **Lots of methods exist to do this...**

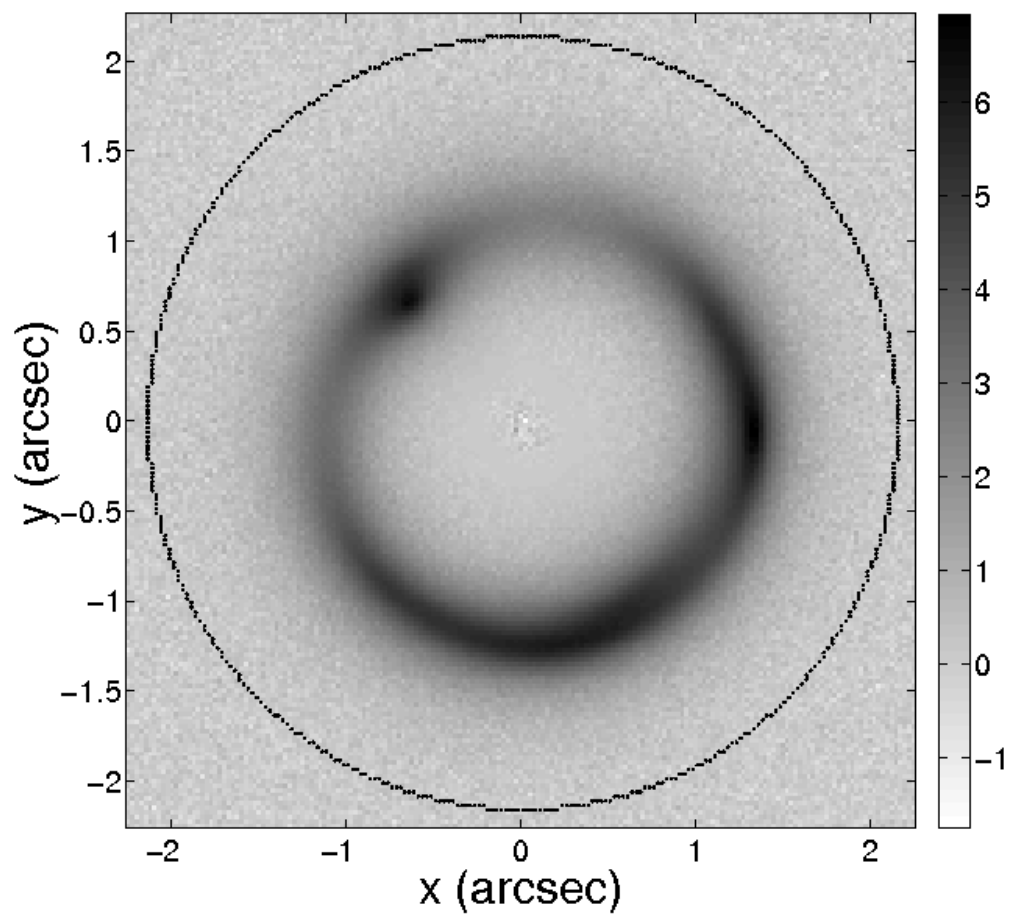
So why AutoLens?

- 1) The Source.
- 2) The Light.
- 3) The Mass.
- 4) The Automation.
- 5) The Science.

***AutoLens:* The Source**

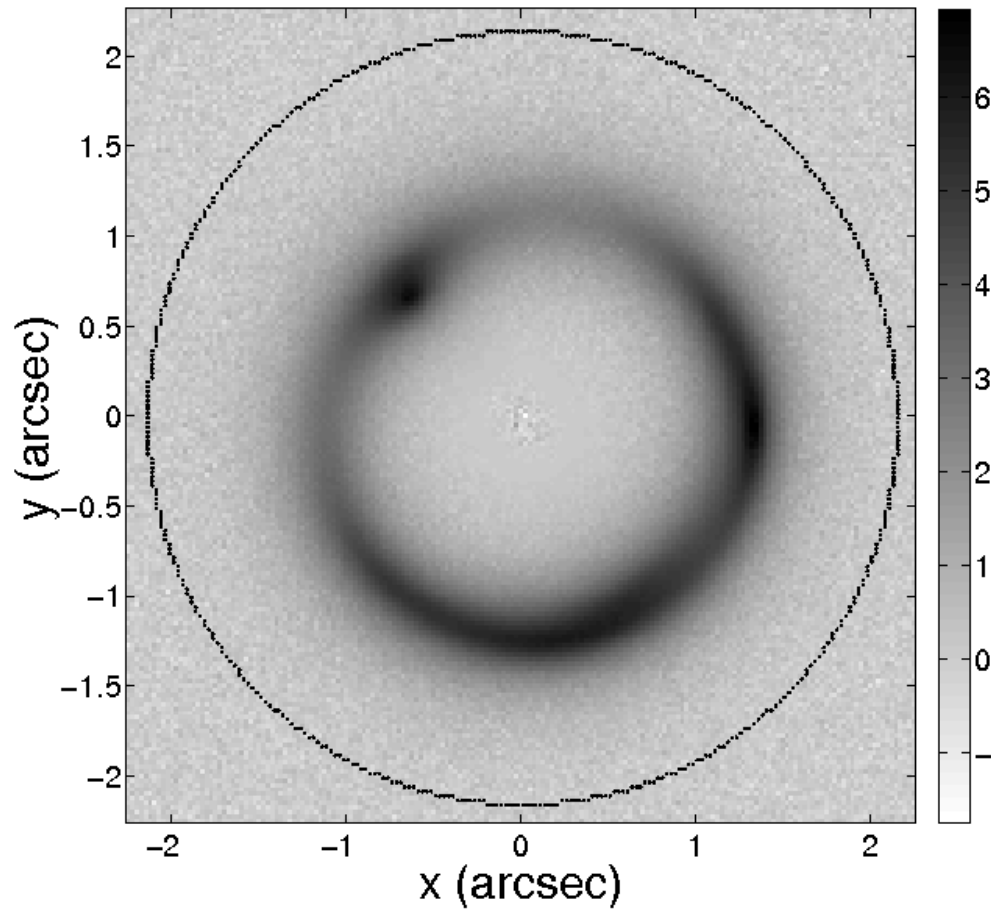
1) Mask Image

Simulated Image

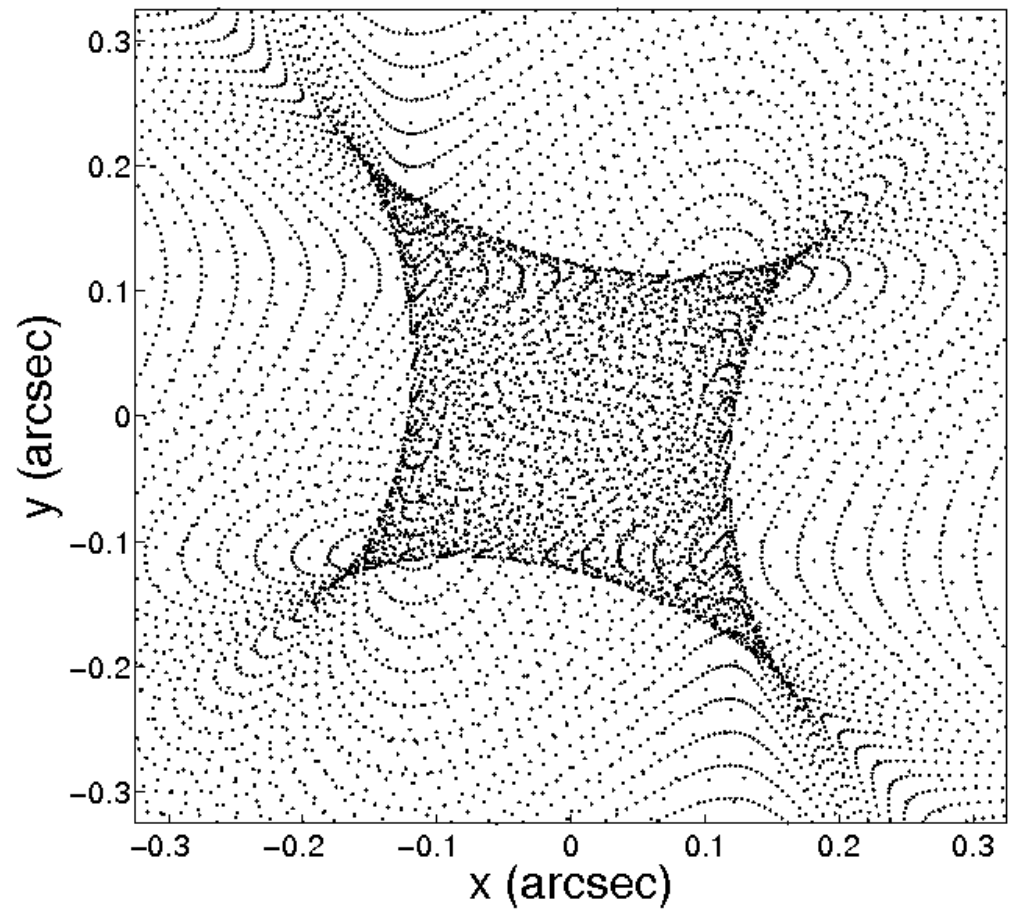


2) Trace Image Pixels to Source Plane

Simulated Image

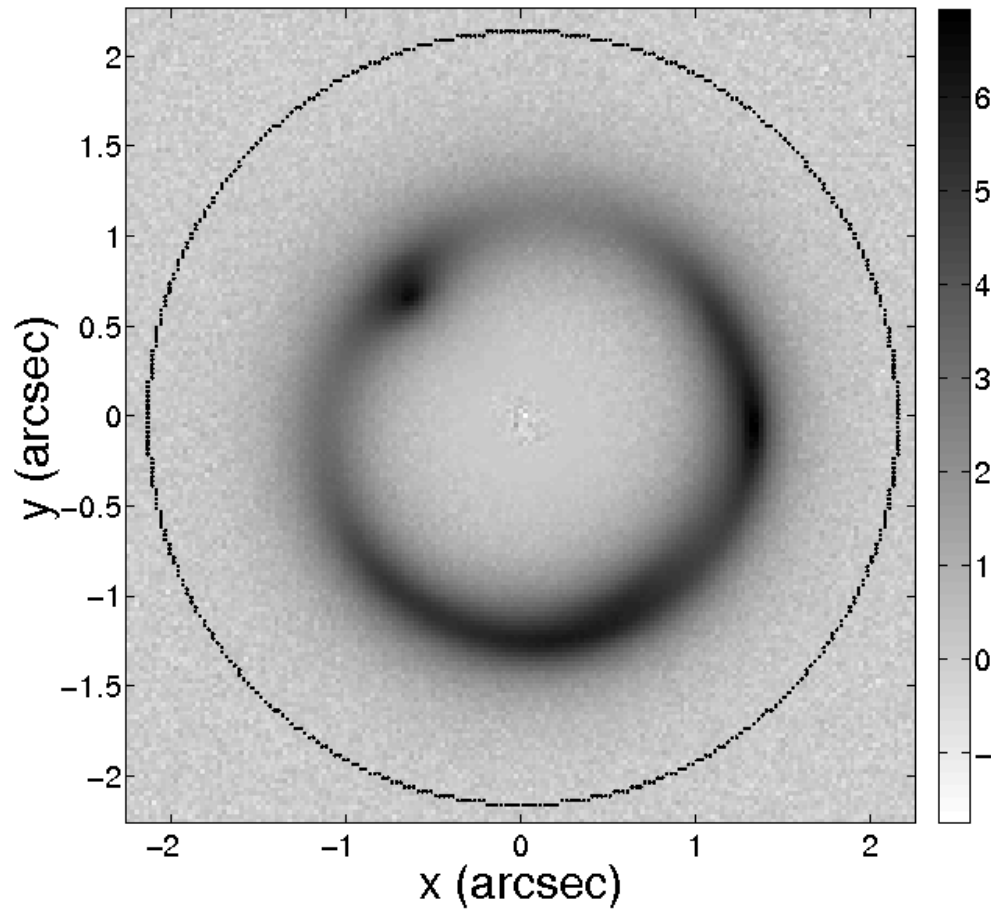


Traced Image Coordinates

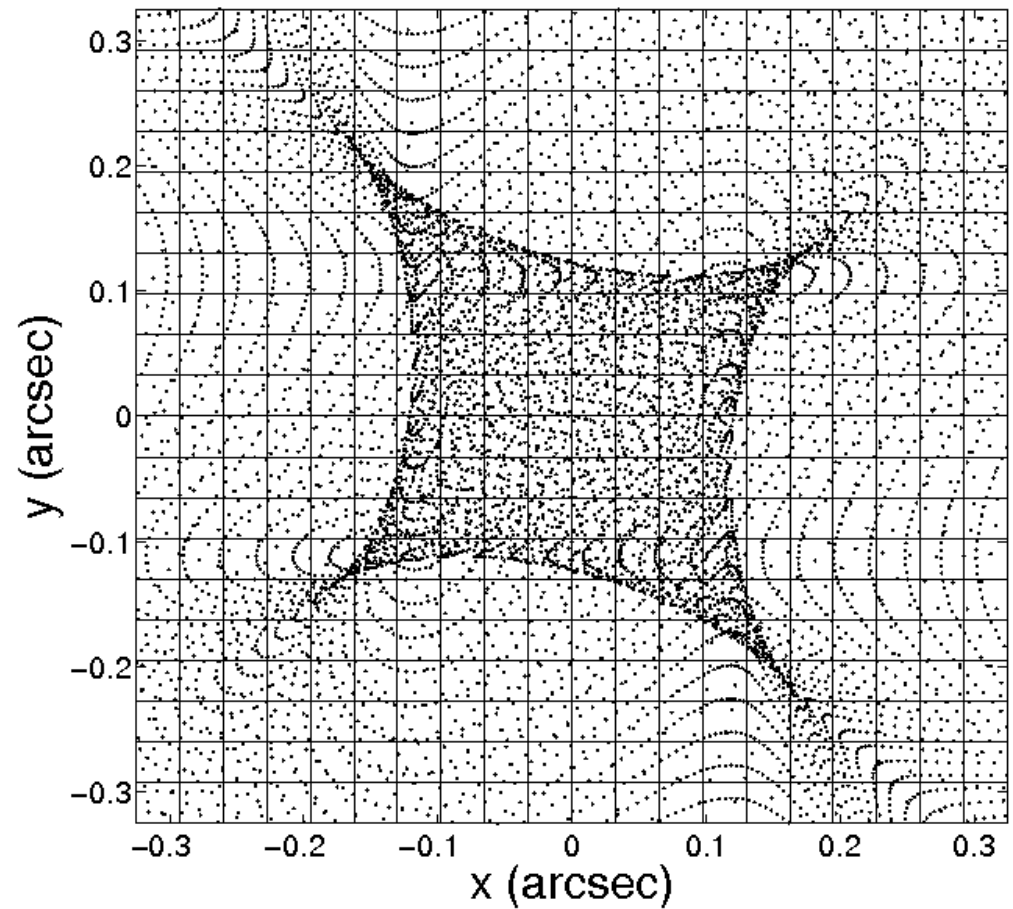


3) Discretize Source Plane

Simulated Image

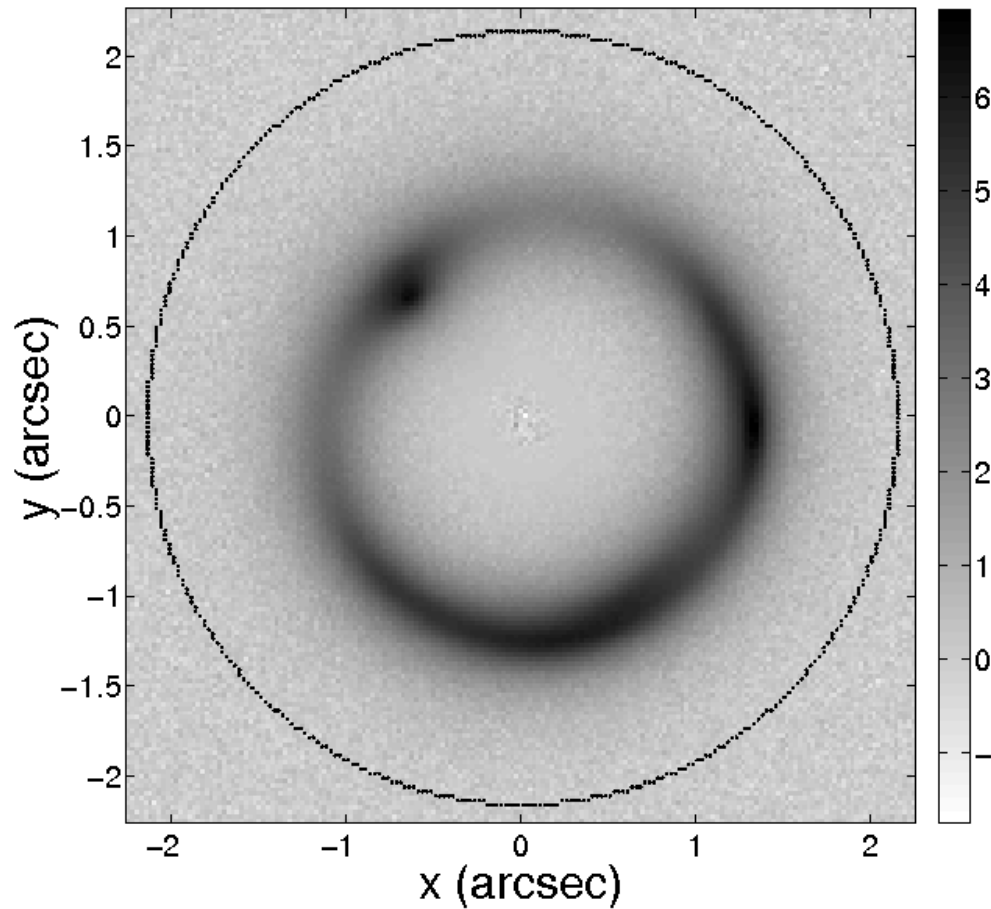


Square Grid Pixelization

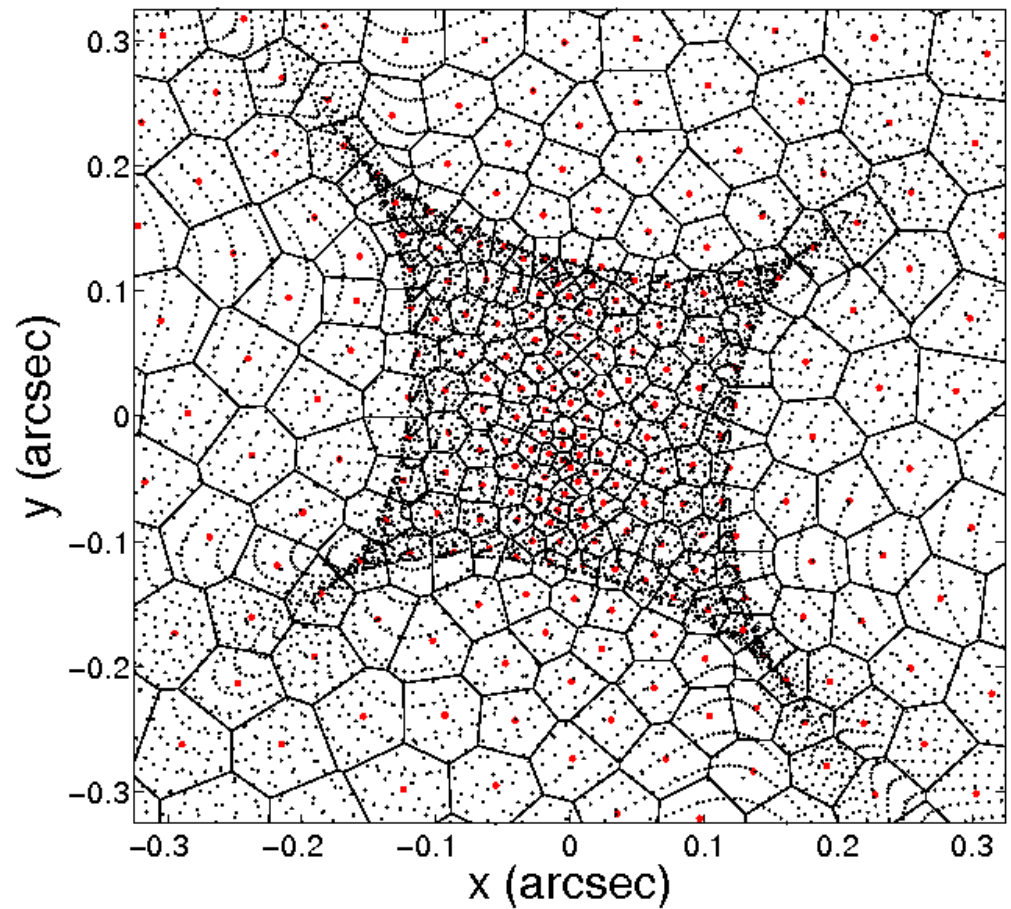


3) Discretize Source Plane

Simulated Image

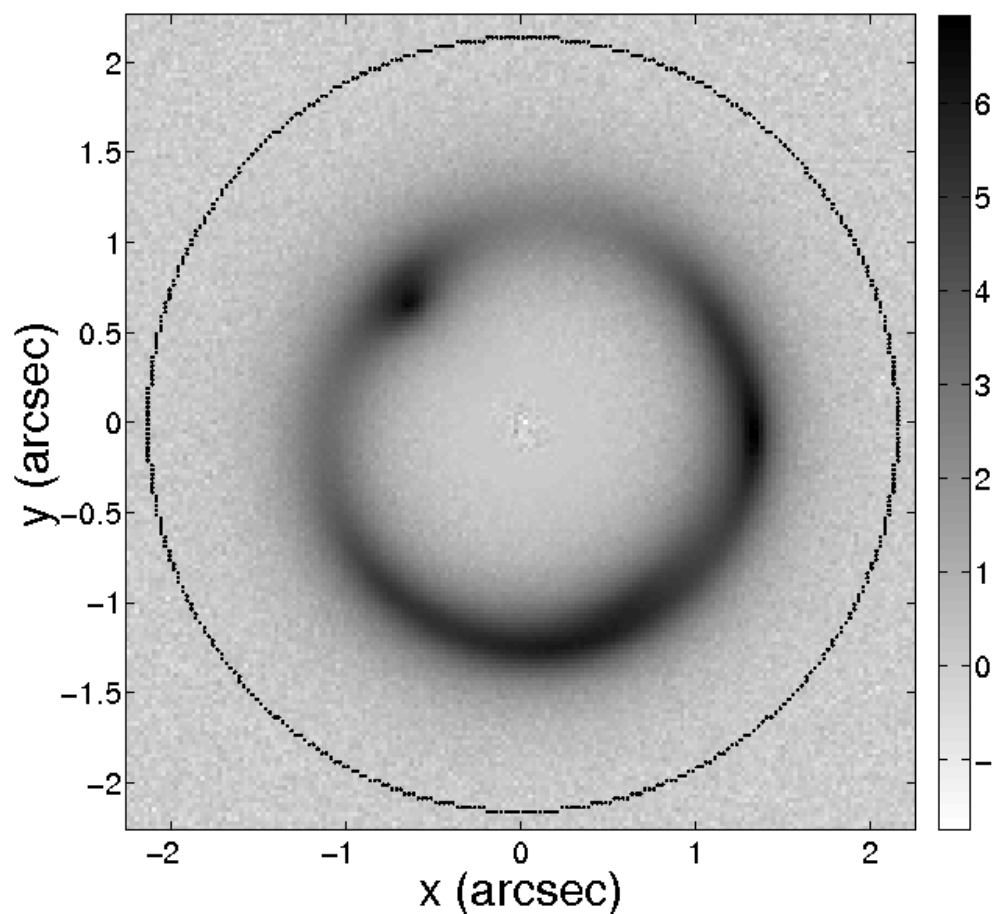


Adaptive Pixelization

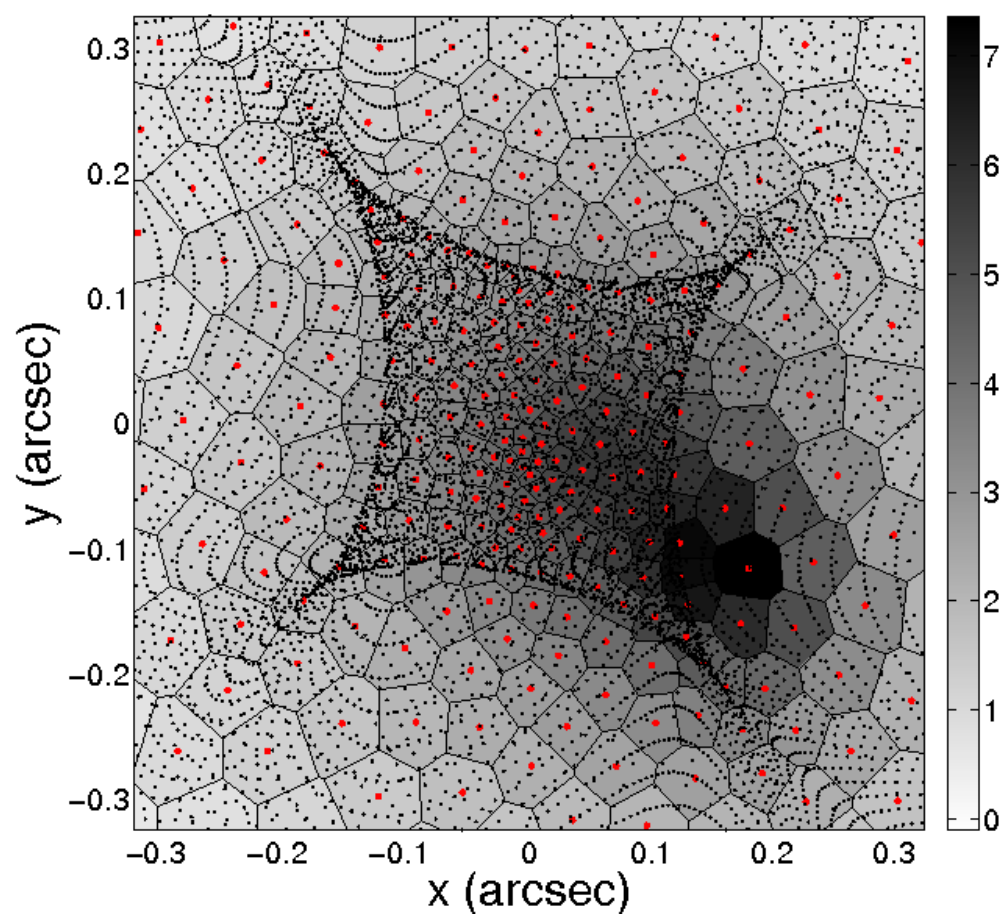


4) Source Reconstruction

Simulated Image

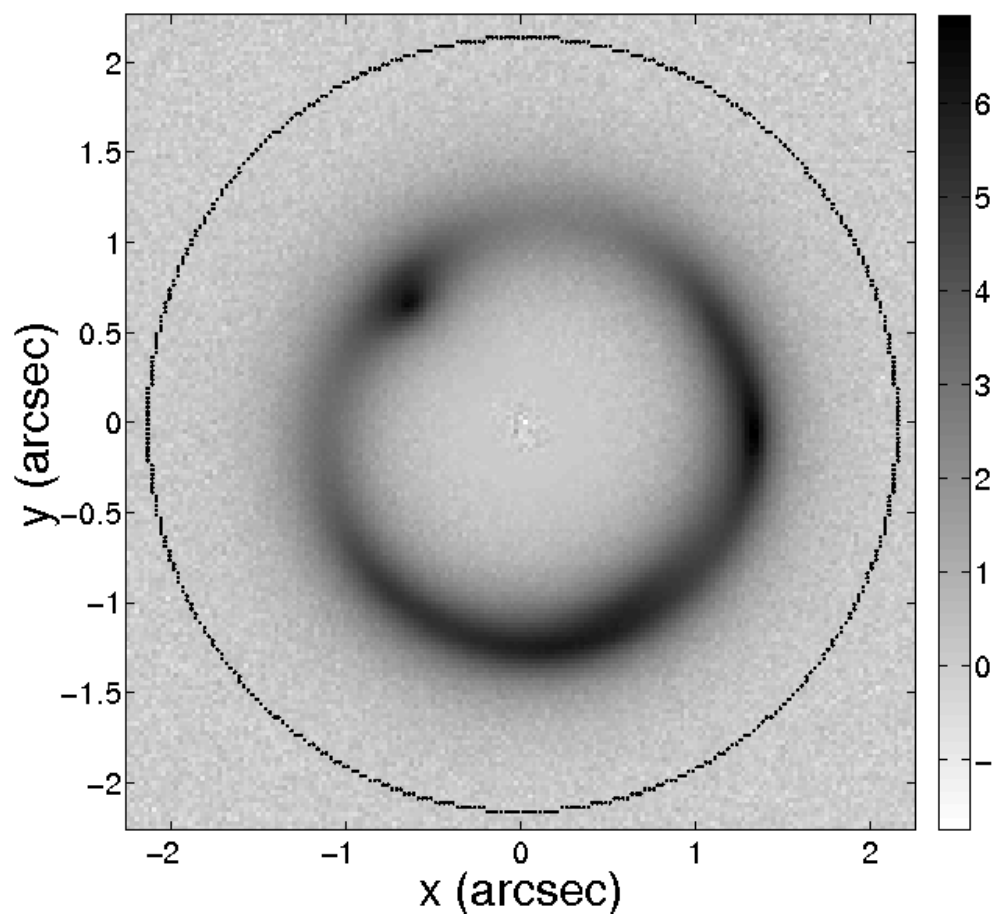


Source Reconstruction

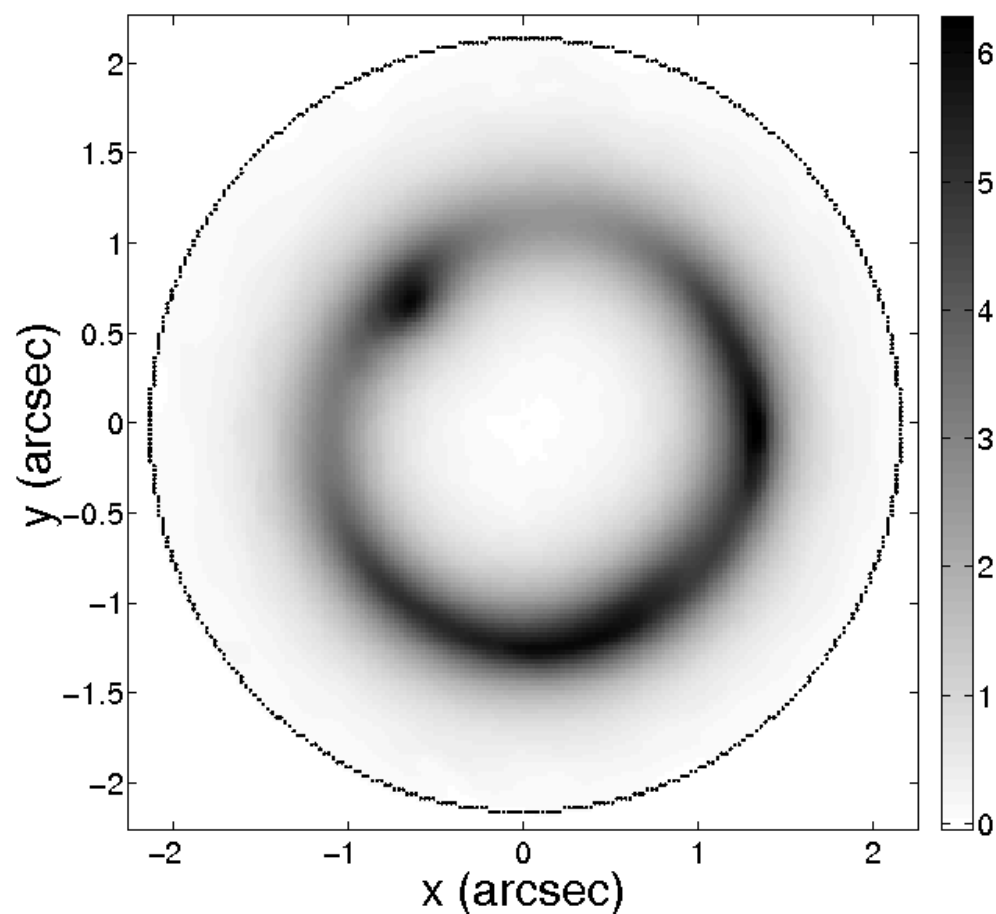


5) Image Reconstruction

Simulated Image



Lensed Source Model

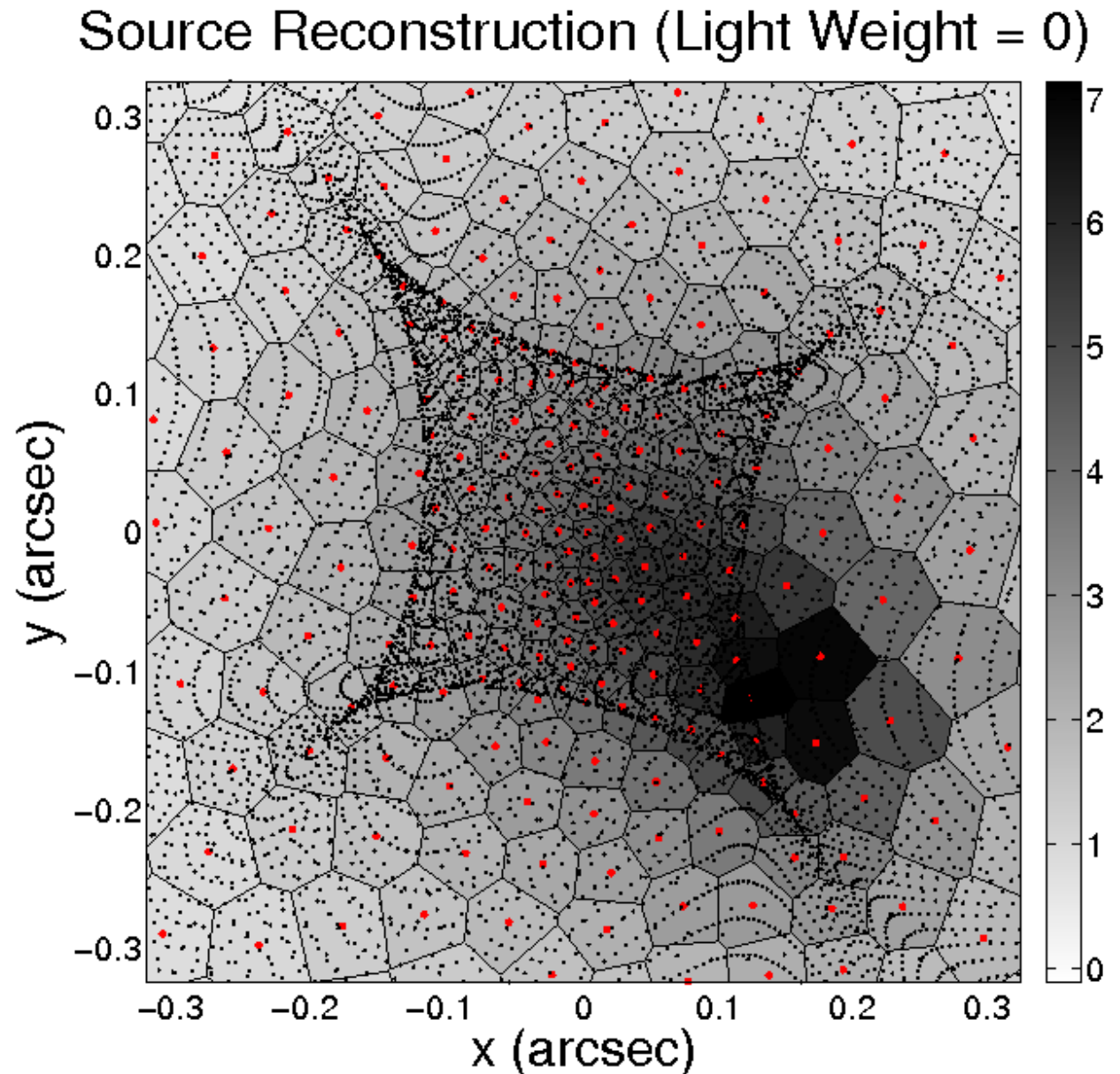


Source Analysis - Details

- **Linear Matrix Inversion of the Lens Equation.**
- **Ill-posed problem, impose linear regularization on source reconstruction:**
 - Prior on the smoothness on the source.
 - Essentially makes sure solution is 'physical'.
- **Self-consistent Bayesian framework (Suyu et al 2006) for source analysis.**
- **Free-form** adaptive grid derived **without geometric prescription.**

Light-Weighted Pixelization

- Evidence Inε
0: -4071.70

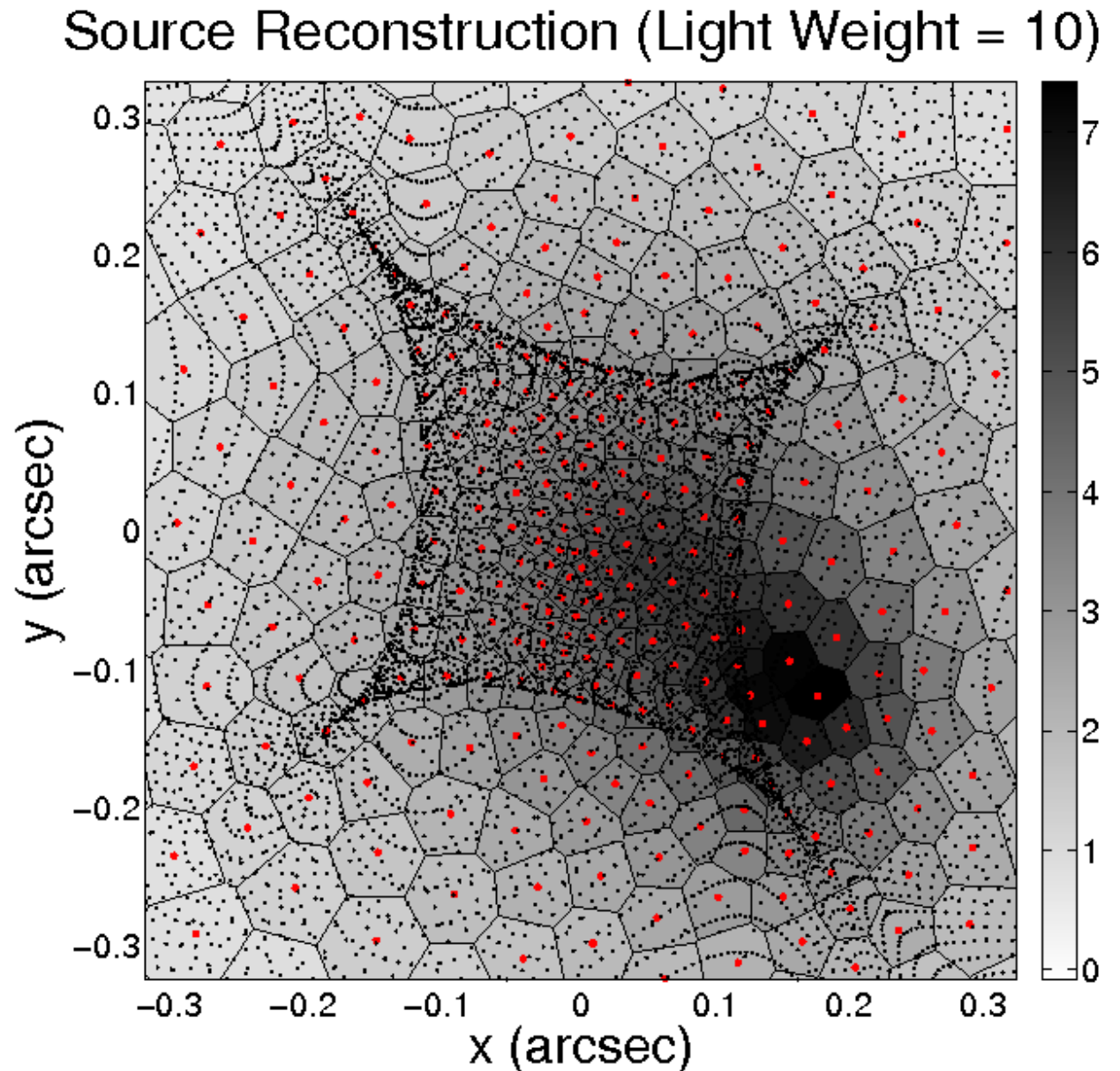


Light-Weighted Pixelization

- Evidence Ine

0: -4071.70

10: -4038.53



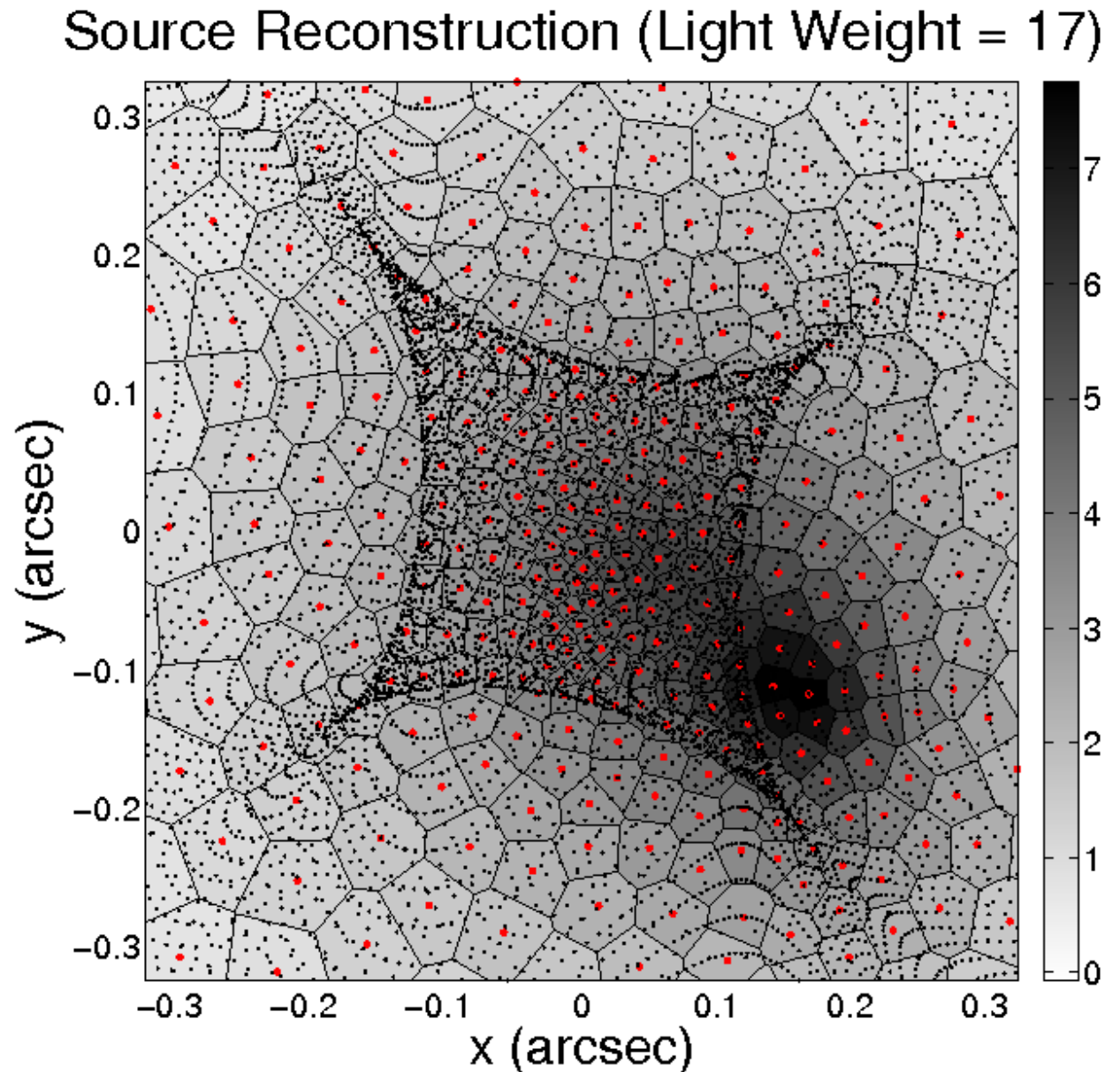
Light-Weighted Pixelization

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17: -4013.76



Light-Weighted Pixelization

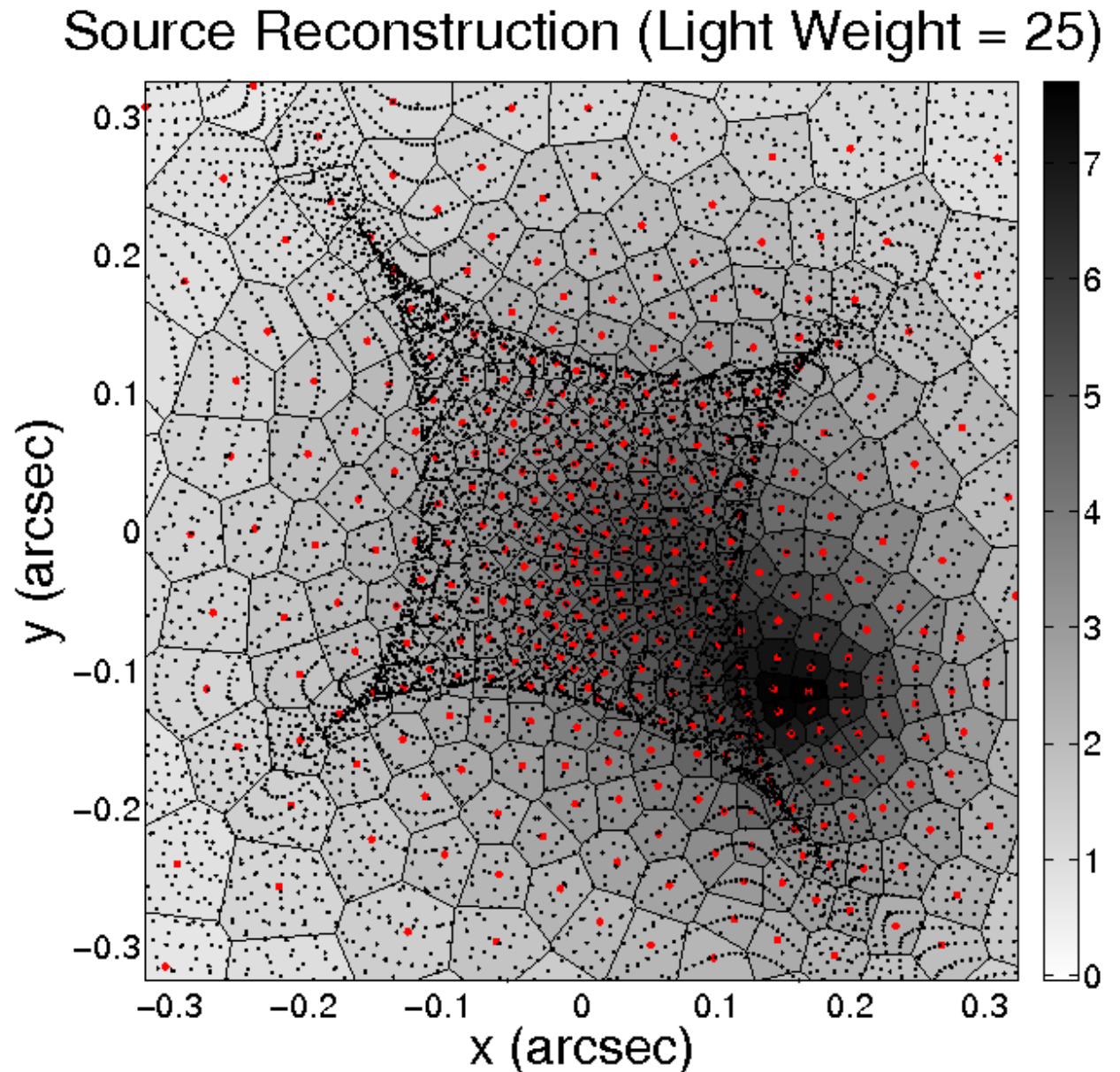
- **Evidence Ine**

0: -4071.70

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25: -4006.76



Light-Weighted Pixelization

- Evidence ϵ

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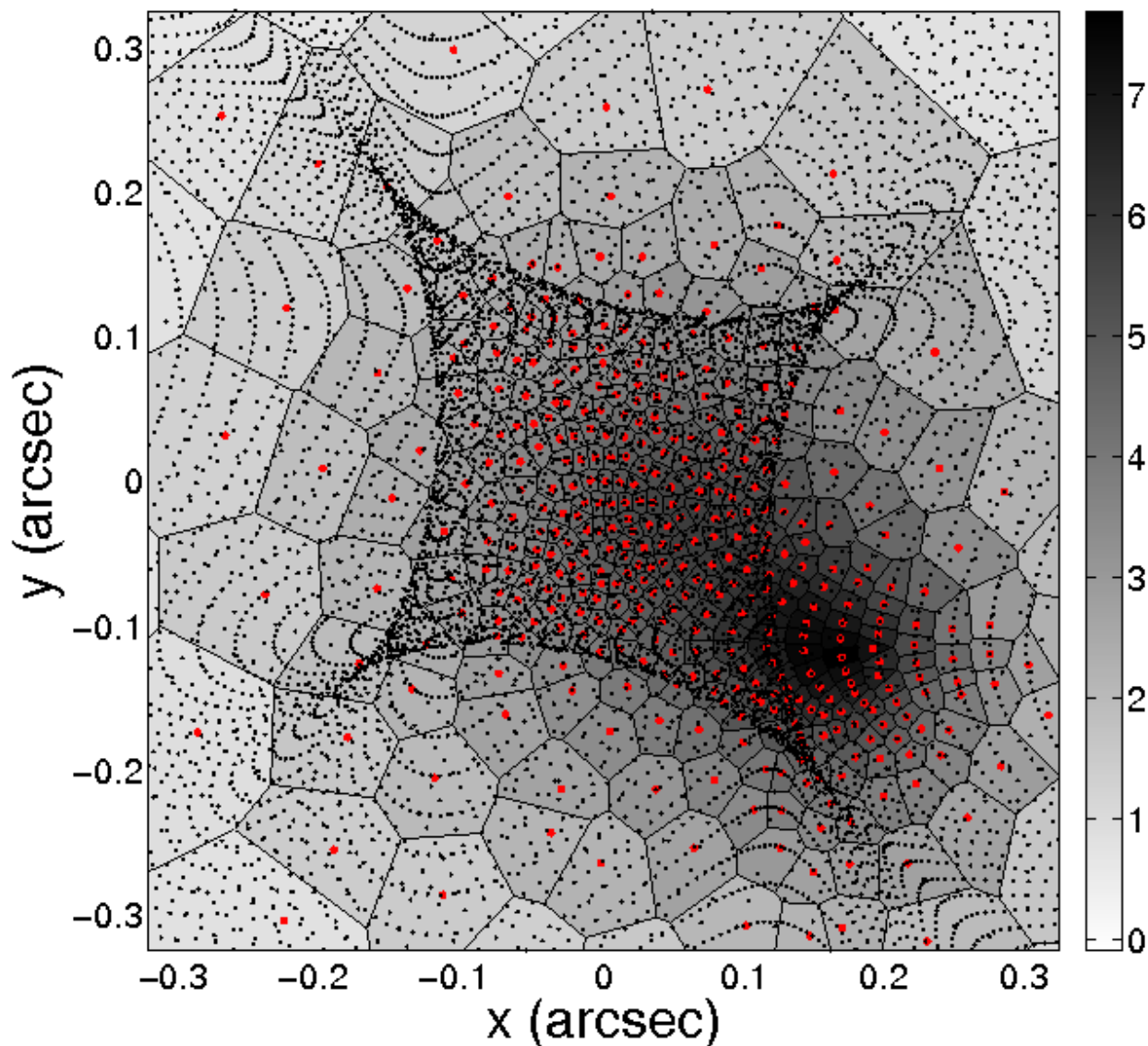
10: -4038.53

17: -4013.76

25: -4006.76

100: -4135.97

Source Reconstruction (Light Weight = 100)



Source Analysis

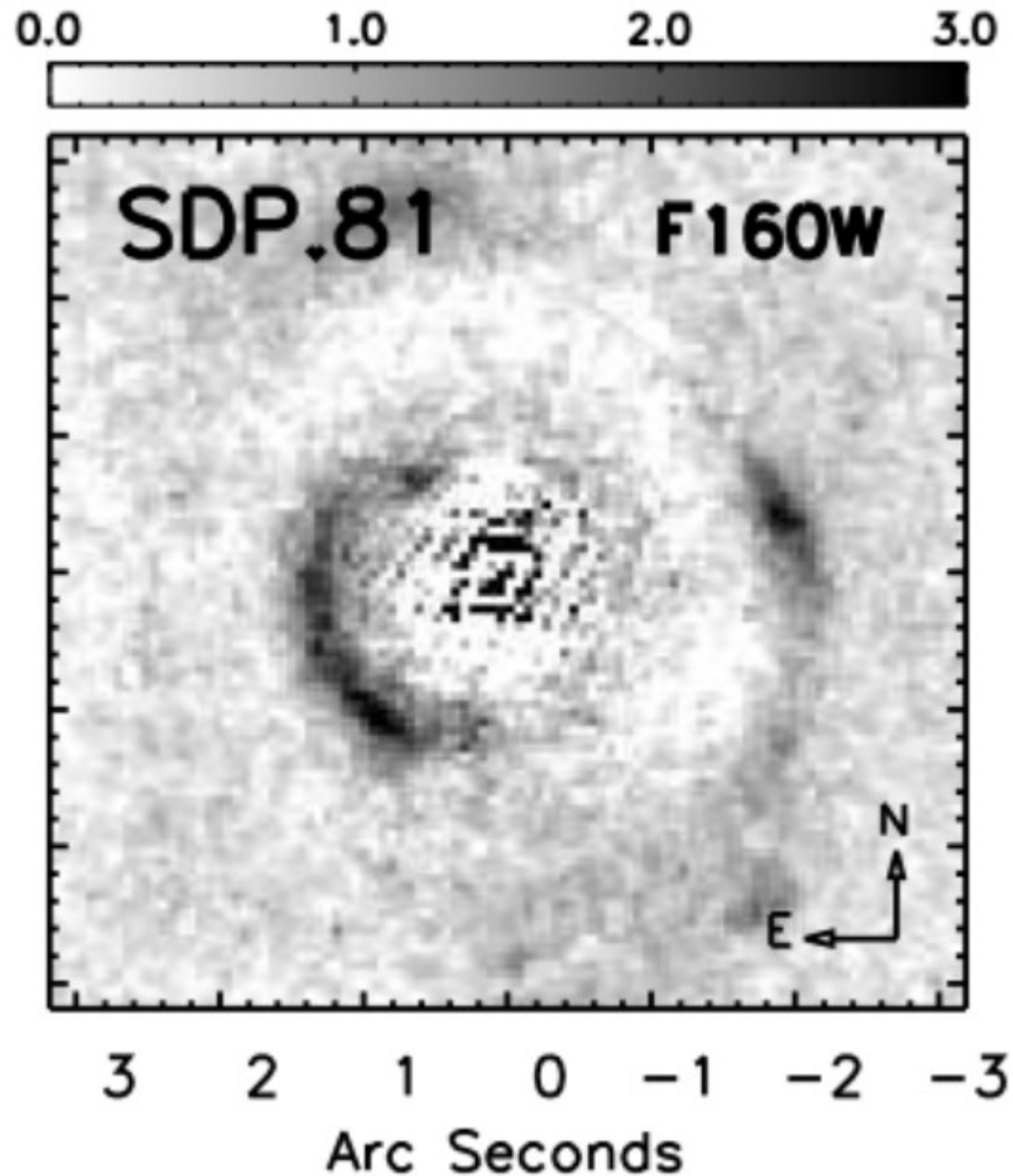
- **Not time to detail all the awesome things *AutoLens*'s source analysis does:**
 - Changing source plane discretization.
 - Luminosity-weighted regularization.
 - Signal processing motivated variance scaling.
- **Speak to me if interested!**

***AutoLens:* Lens Light**

The Lens Light

Negrello et al.
2014

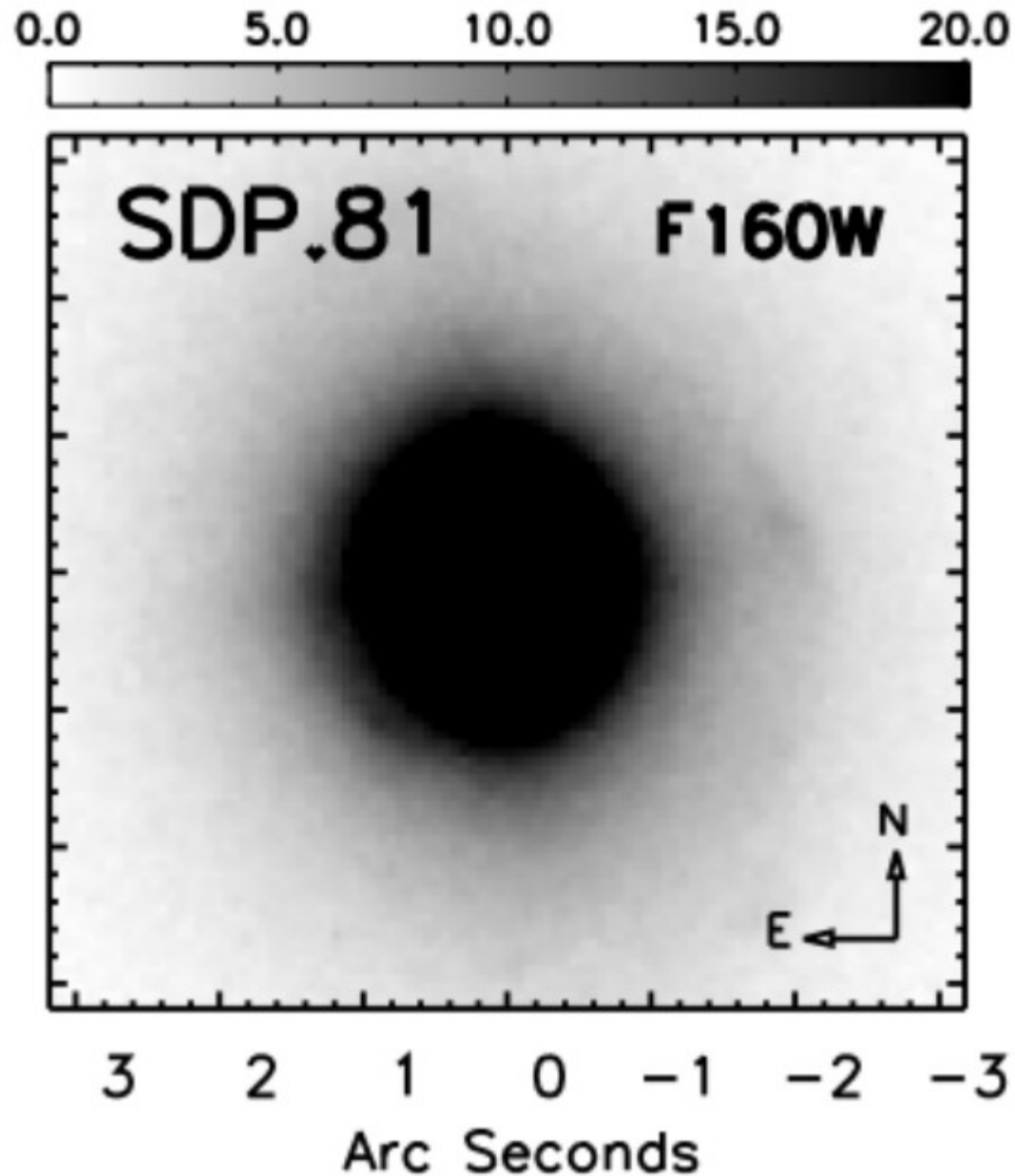
Dye et al.
2014



The Lens Light

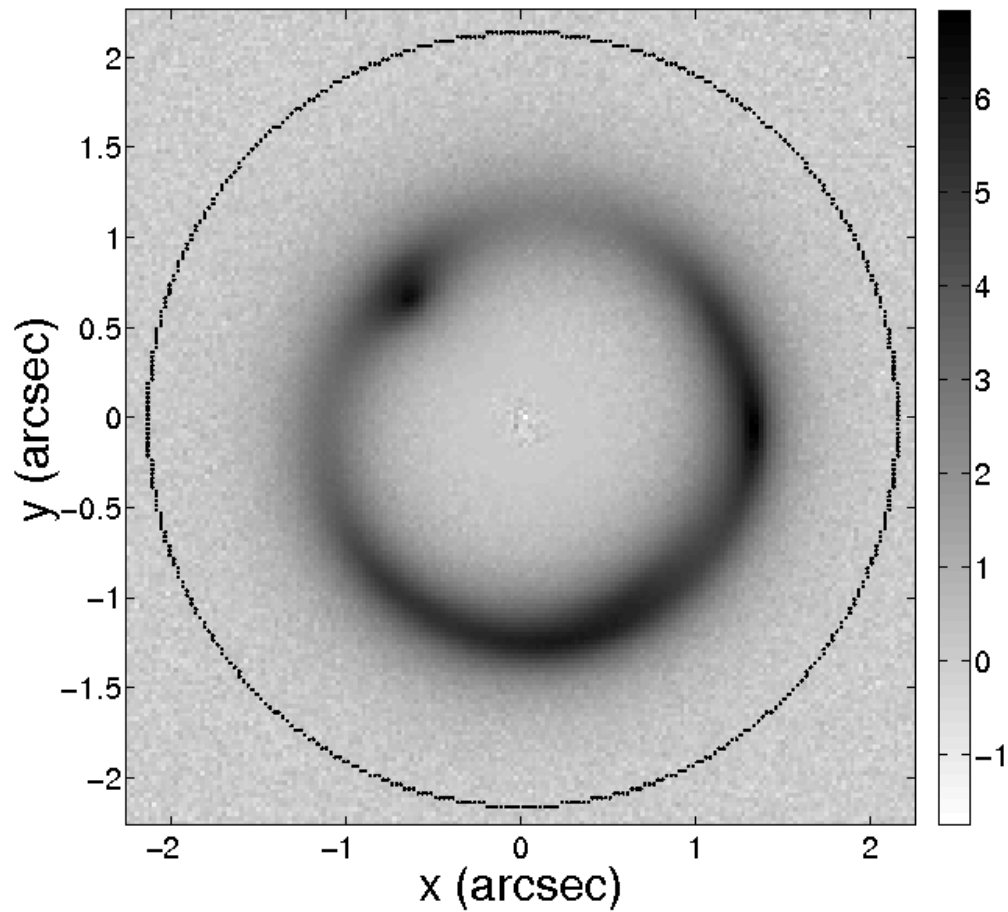
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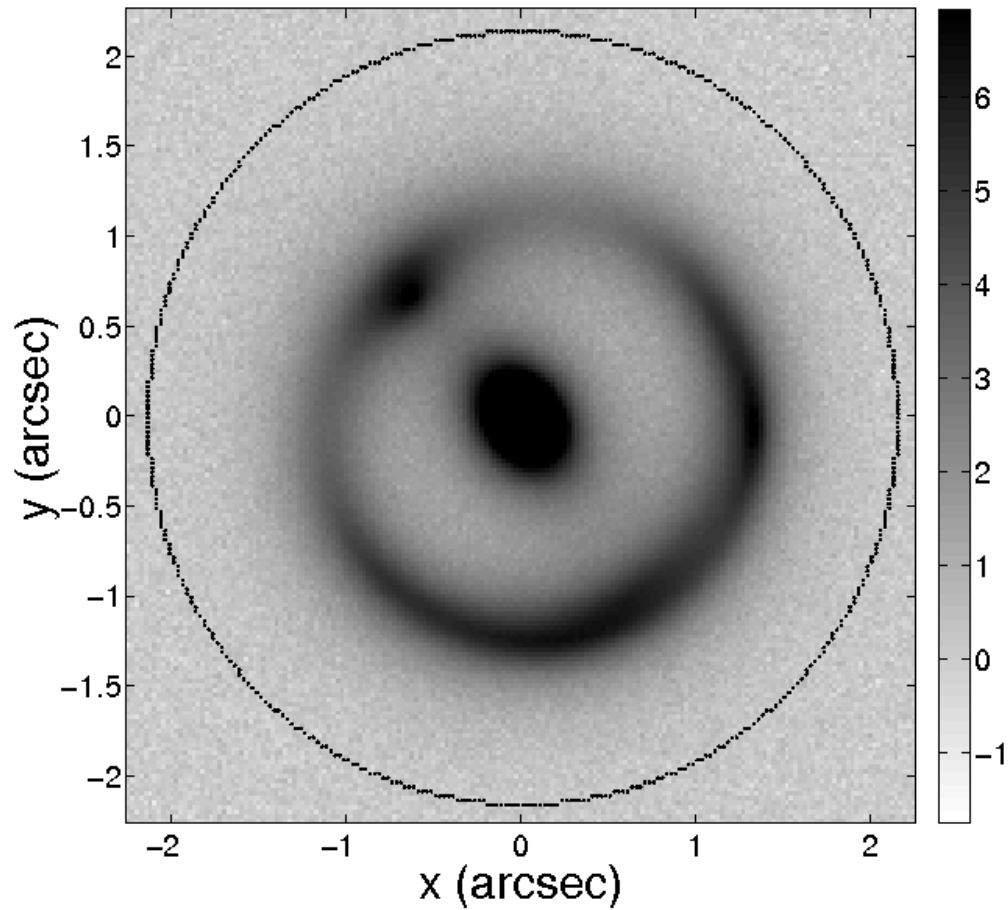
Lens Light Modeling

Simulated Image



Lens Light Modeling

Simulated Image



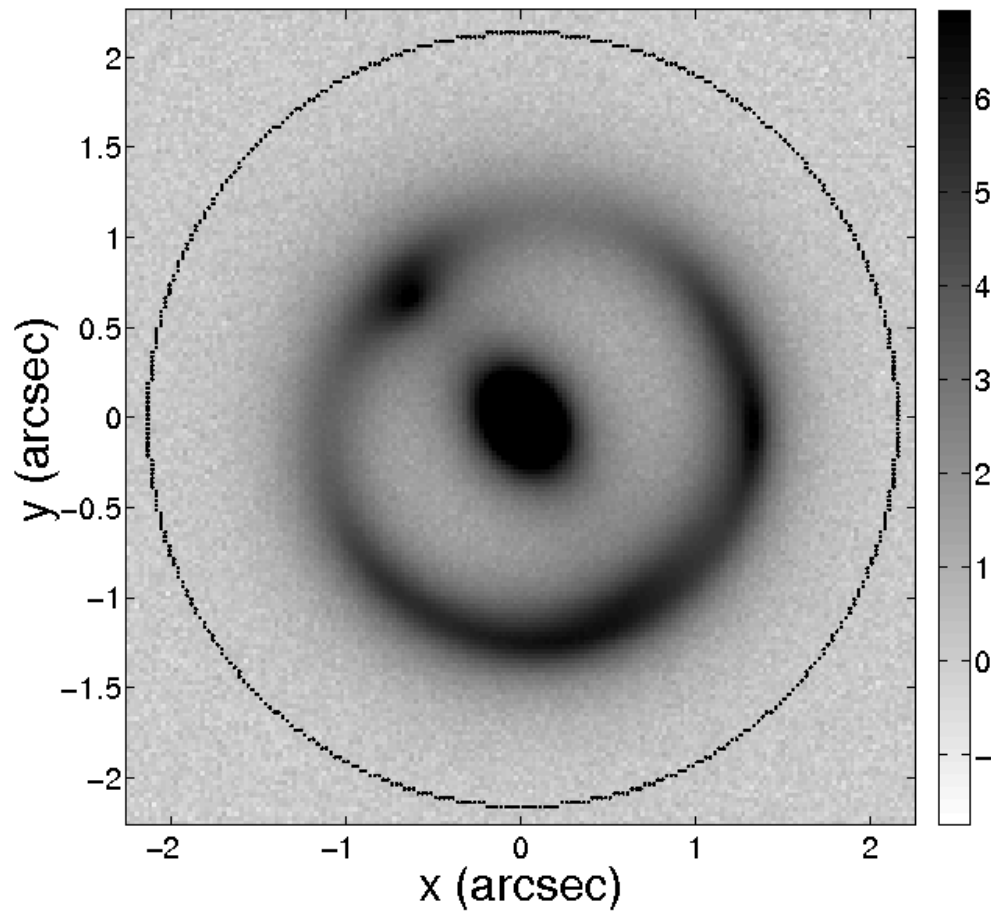
Building a Light Profile

- Use elliptical Sersic profile:

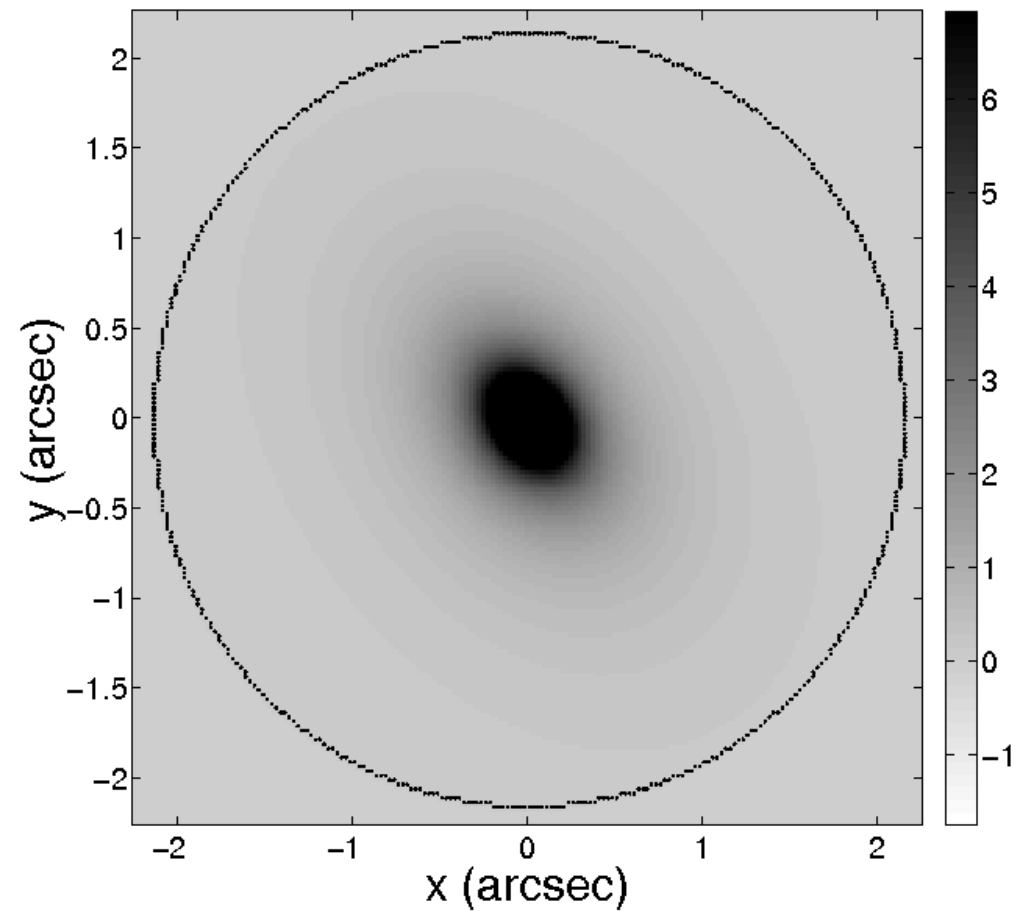
$$I_{Sersic}(\xi_l) = I_l \exp \left\{ -k_l \left[\left(\frac{\xi_l}{R_l} \right)^{\frac{1}{n_l}} - 1 \right] \right\}$$

Lens Light Modeling - Sersic

Simulated Image

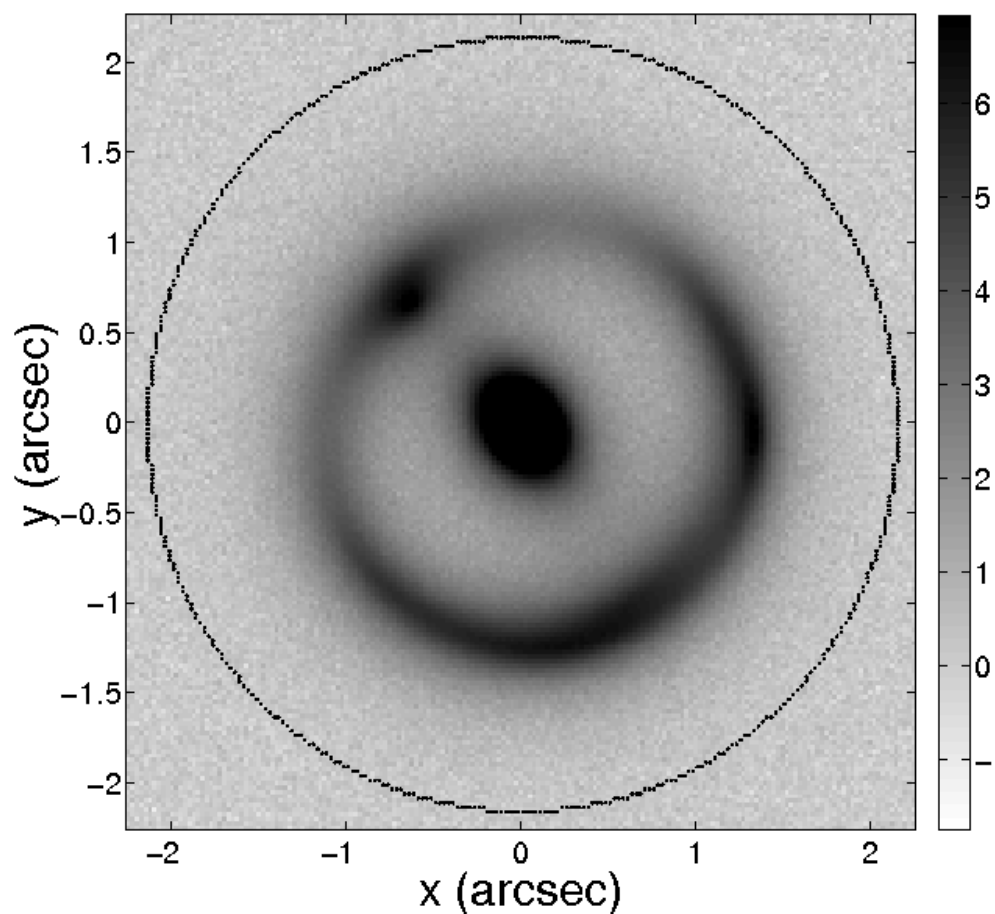


Adaptive SLI Lens Light Model

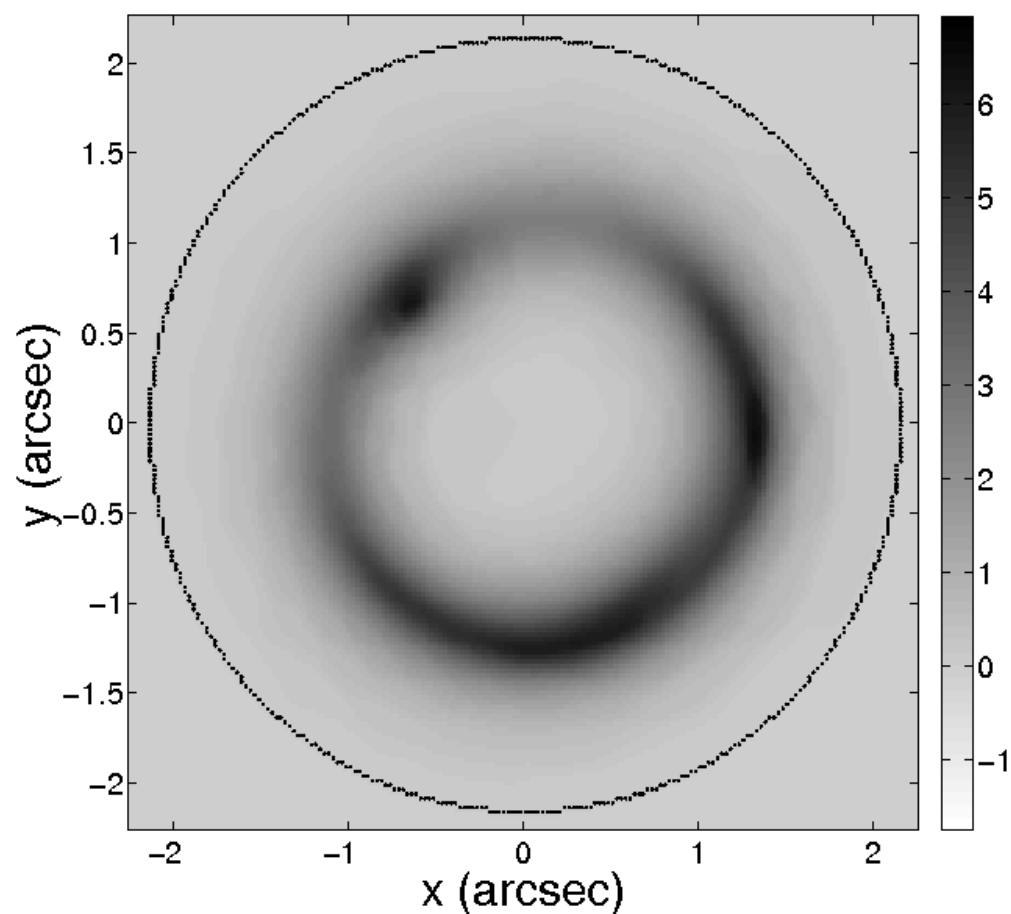


Lens Light Modeling - Sersic

Simulated Image

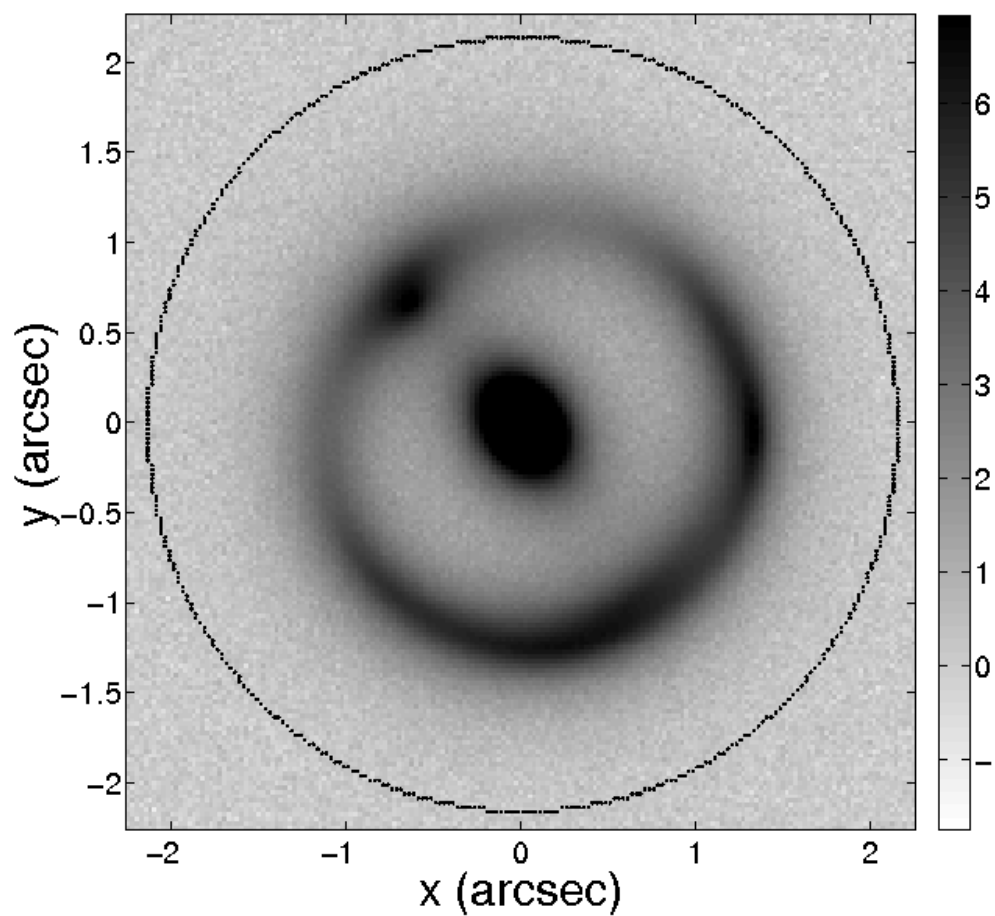


Adaptive SLI Source Model

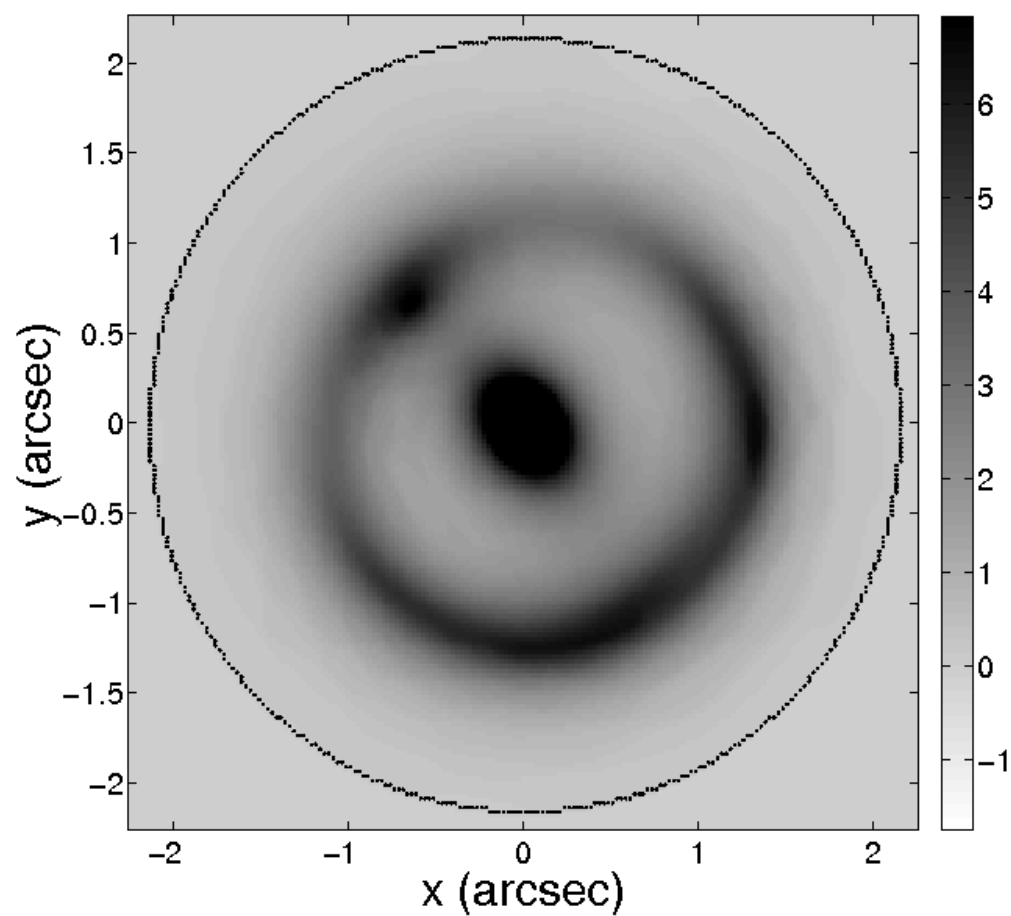


Lens Light Modeling - Sersic

Simulated Image



Adaptive SLI Lens + Source Model



Light Profile

- **Use elliptical Sersic profile:**

$$I_{Sersic}(\xi_l) = I_l \exp \left\{ -k_l \left[\left(\frac{\xi_l}{R_l} \right)^{\frac{1}{n_l}} - 1 \right] \right\}$$

- ... Or multiple Sersic's (or de Vaucouleurs / Exponentials / Gaussians / etc.)
- Choose light model with **Bayesian model comparison.**

Lens Light Modeling

- Clean separation of lens and source light.
- Infer the lens's light profile, Bulge-to-Disk ratio, etc.
- **Astronomers have been studying galaxy light profiles for over half a century.**
- **Taken fifteen years of strong lens modeling to measure one!**

***AutoLens:* Lens Mass**

Mass Modeling – Non-linear Analysis

- **Standard non-linear sampling problem.**
- **Use nested sampling with *MultiNest* throughout *AutoLens*.**
- **Mass model also chosen via Bayesian model comparison:**
 - e.g. is an external shear required?

Mass Model

- Strong lens modeling historically uses an elliptical power-law density profile representing the **total** mass:

$$\rho(r) = \rho_o (r/r_o)^{-\alpha}$$

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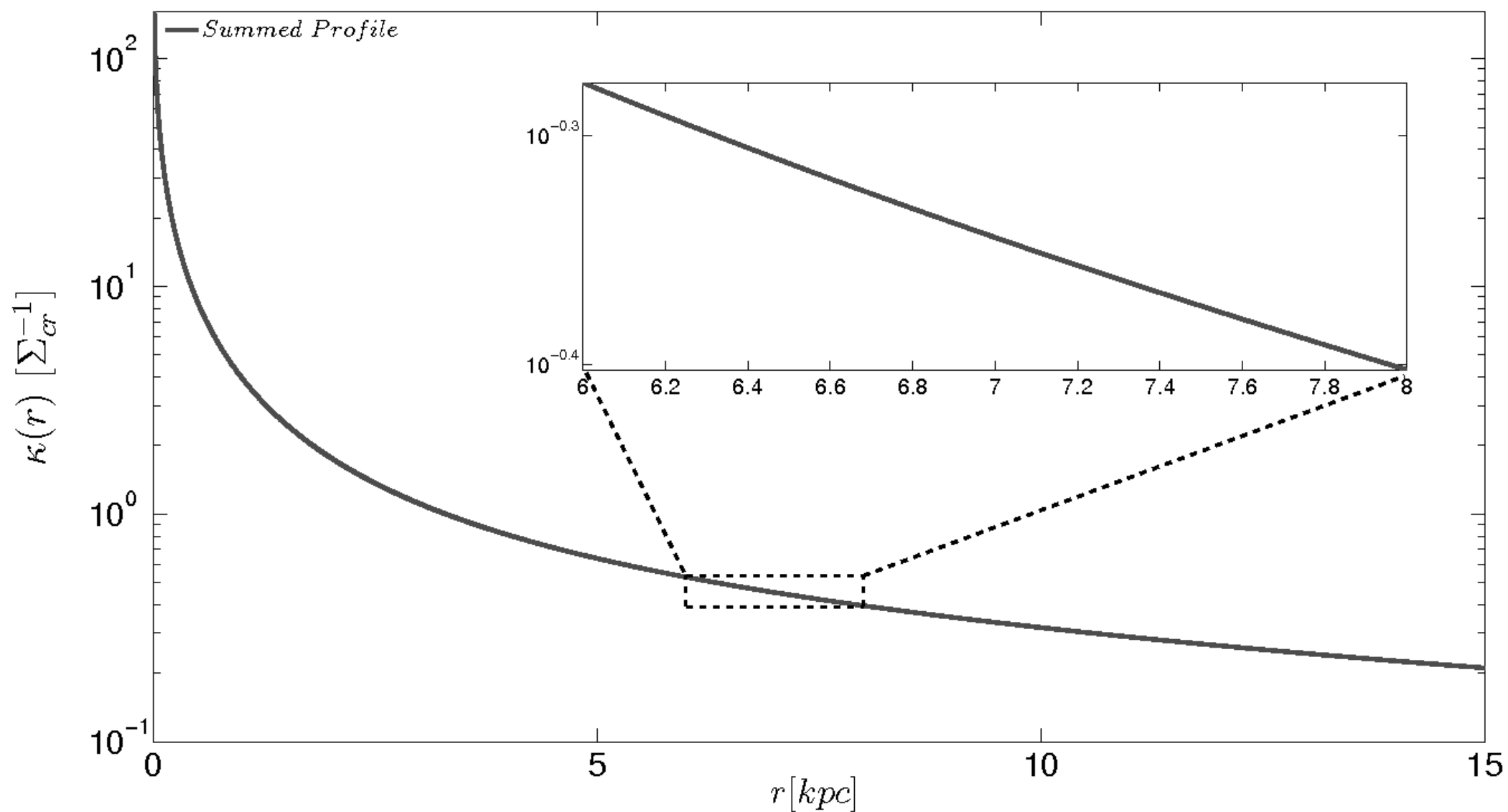
- Now advocate **decomposed mass modeling** representing the **light** (Multi-Sersic) ...

$$\kappa_{Sersic}(\xi_l) = \Psi_l I_l \exp \left\{ -k_l \left[\left(\frac{\xi_l}{R_l} \right)^{\frac{1}{n_l}} - 1 \right] \right\}$$

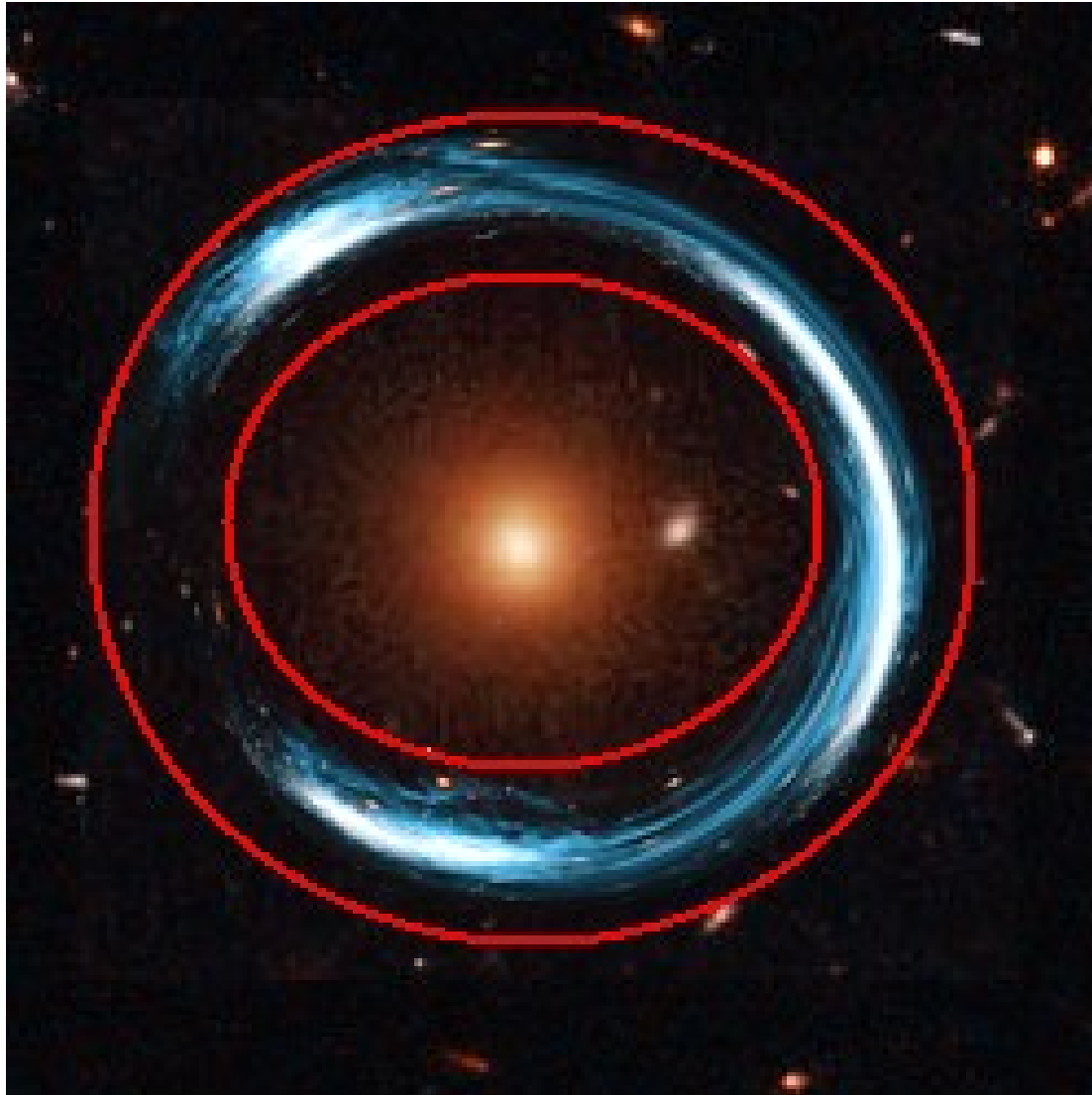
- ... and **dark** (gNFW) matter:

$$\rho = \frac{\rho_s}{(r/r_s)^\gamma (1 + r/r_s)^{3-\gamma}} \quad \begin{array}{l} r_s = 50 \text{ kpc} \\ \text{(Bullock et al. 2001)} \end{array}$$

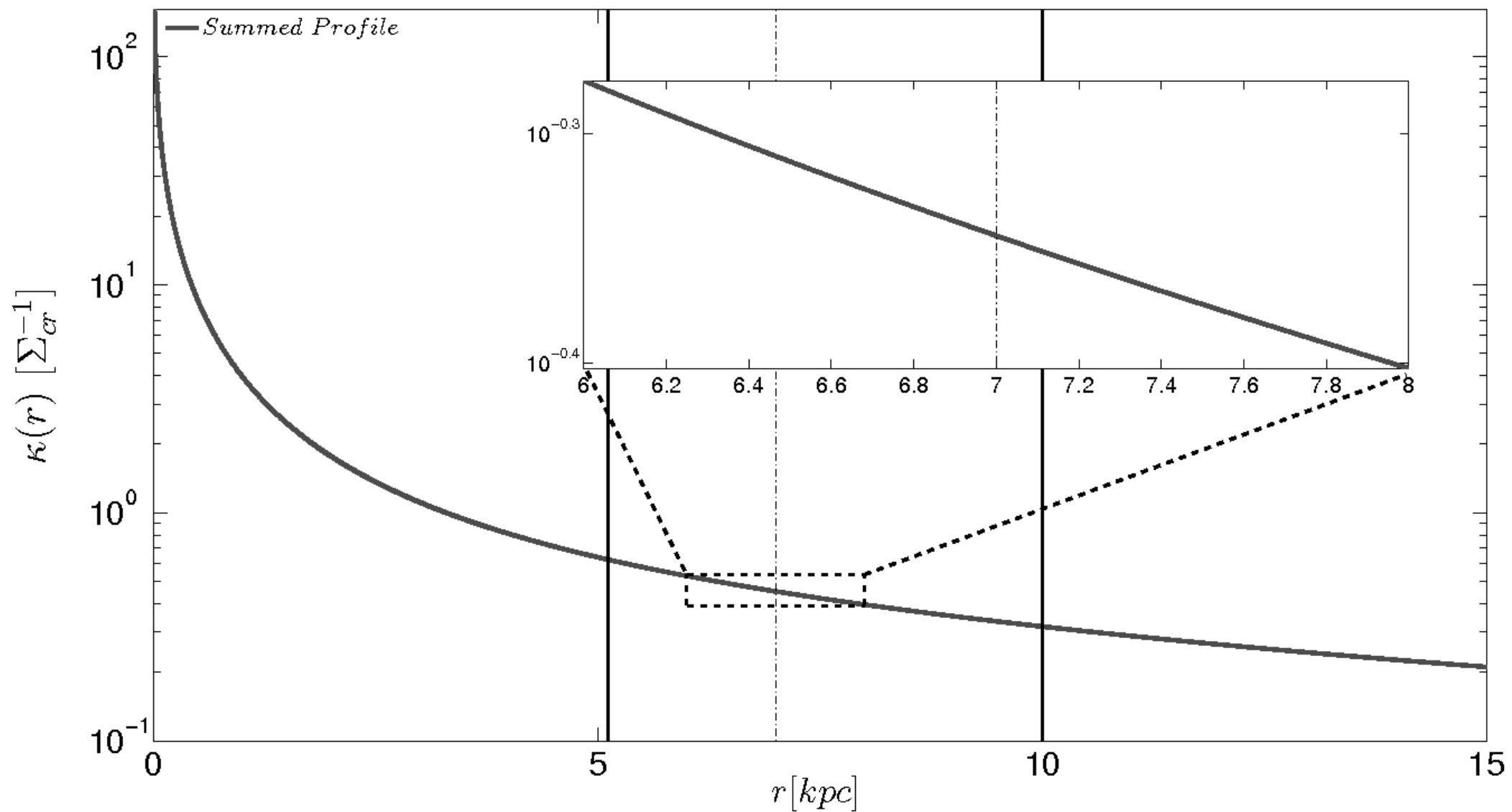
Mass Modeling – Total Mass



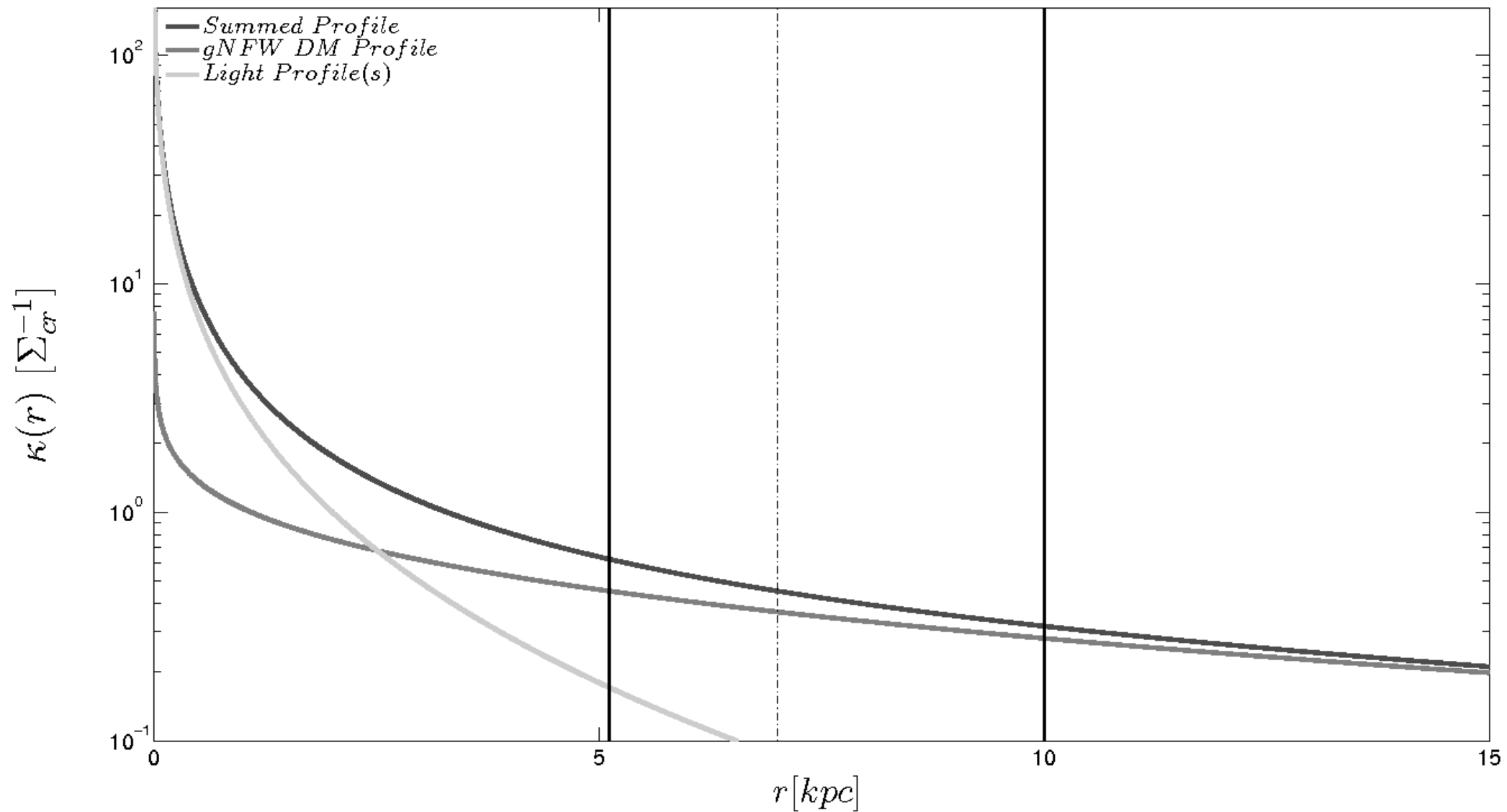
What does a strong lens tell us?



Mass Modeling – R_{Ein}



Mass Modeling – Light / Dark Matter



Mass Modeling – Non-Axisymmetry

- **2d projected geometry of the light and dark matter can be modeled independently.**
 - Different **ellipticities**?
 - **Rotational** (mis)alignment?
 - **Positional** (mis)alignment?
- **Again choose model via Bayesian Model Comparison.**

Automated Analysis

Why **AutoLens**?

- **S Dye** - 2005 - 2015 – 9 lenses.
- **S Suyu** – 2006 – 2015 – 4 lenses
- **S Vegetti** – 2009 – 2015 – 13 lenses
- **M Barnabe** – 2007 – 2015 – 16 lenses
- **Total: ~50**

Why **AutoLens**?

- There are **over 250** HST images in the HST archive that warrant this analysis.
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- All previous studies ignored the lens light.
- 50 lenses since 2005.
- 5 lenses a year.
- 100 000 thousand lenses.

Why **AutoLens**?

- There are **over 250** HST images in the HST archive that warrant this analysis.
- All previous studies ignored the lens light.
- **20 000 years to model the Euclid sample...**

Why so slow?

- 14-20 non-linear lens parameters with up to 7+ way degeneracies.
- 200-800+ linear source pixel parameters, with random discretization.
- 10+ hyper-parameters controlling Bayesian inference, requiring independent optimization.
- And lots more... (PSF blurring, Over-sampling, Image Masking, etc.)

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- 10+ hyper-parameters controlling Bayesian inference, requiring independent optimization.
- And lots more... (PSF blurring, Over sampling, Image Masking, etc.)
- **Previous methods control all of this 'by hand'!**

Automated Analysis

- **All results generated with *AutoLens* are automated - require no user-intervention!**
- **Rigorously tested *AutoLens* on 20 simulated images, testing different:**
 - Source and Lens Morphologies.
 - Lens Mass Profiles.
 - Lensing Geometries.
- **Will be shown in two upcoming publications.**

ER-0047-2808

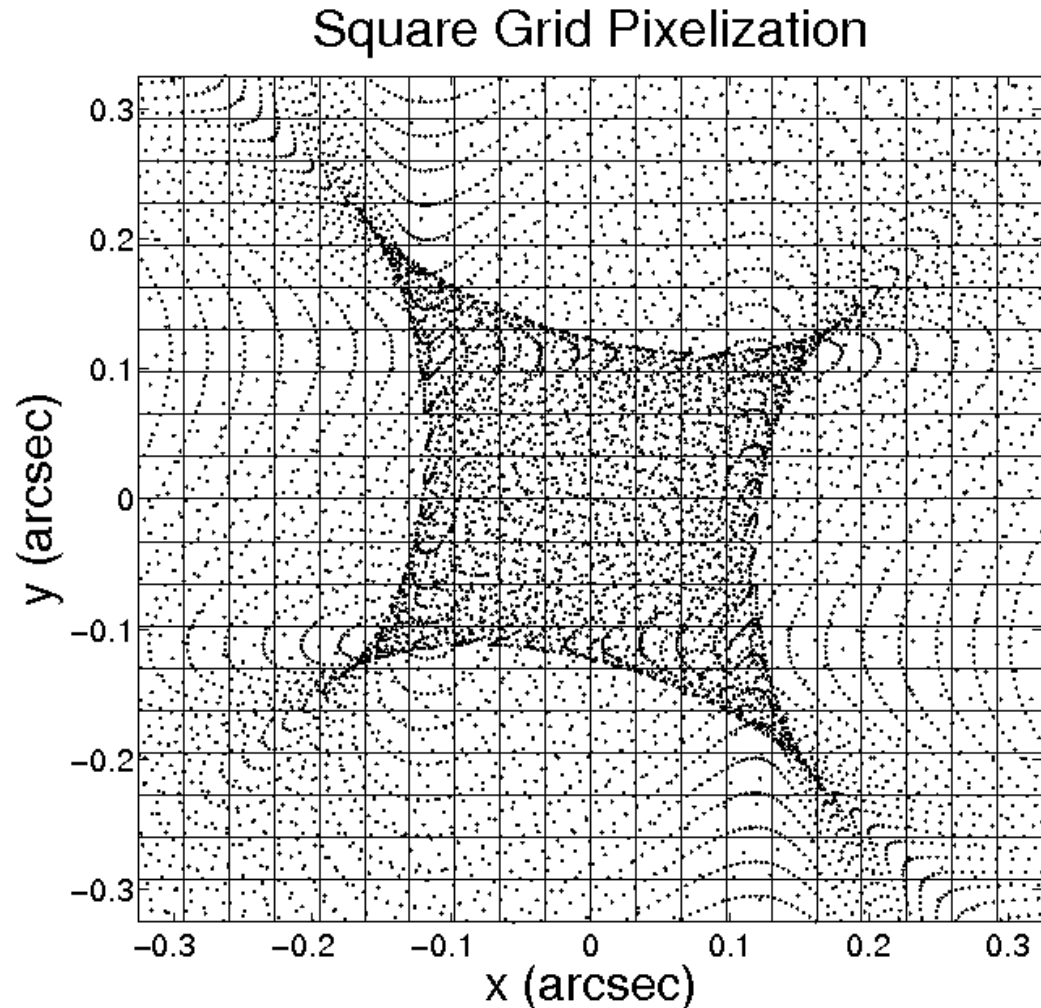
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Euclid

Extra Slides

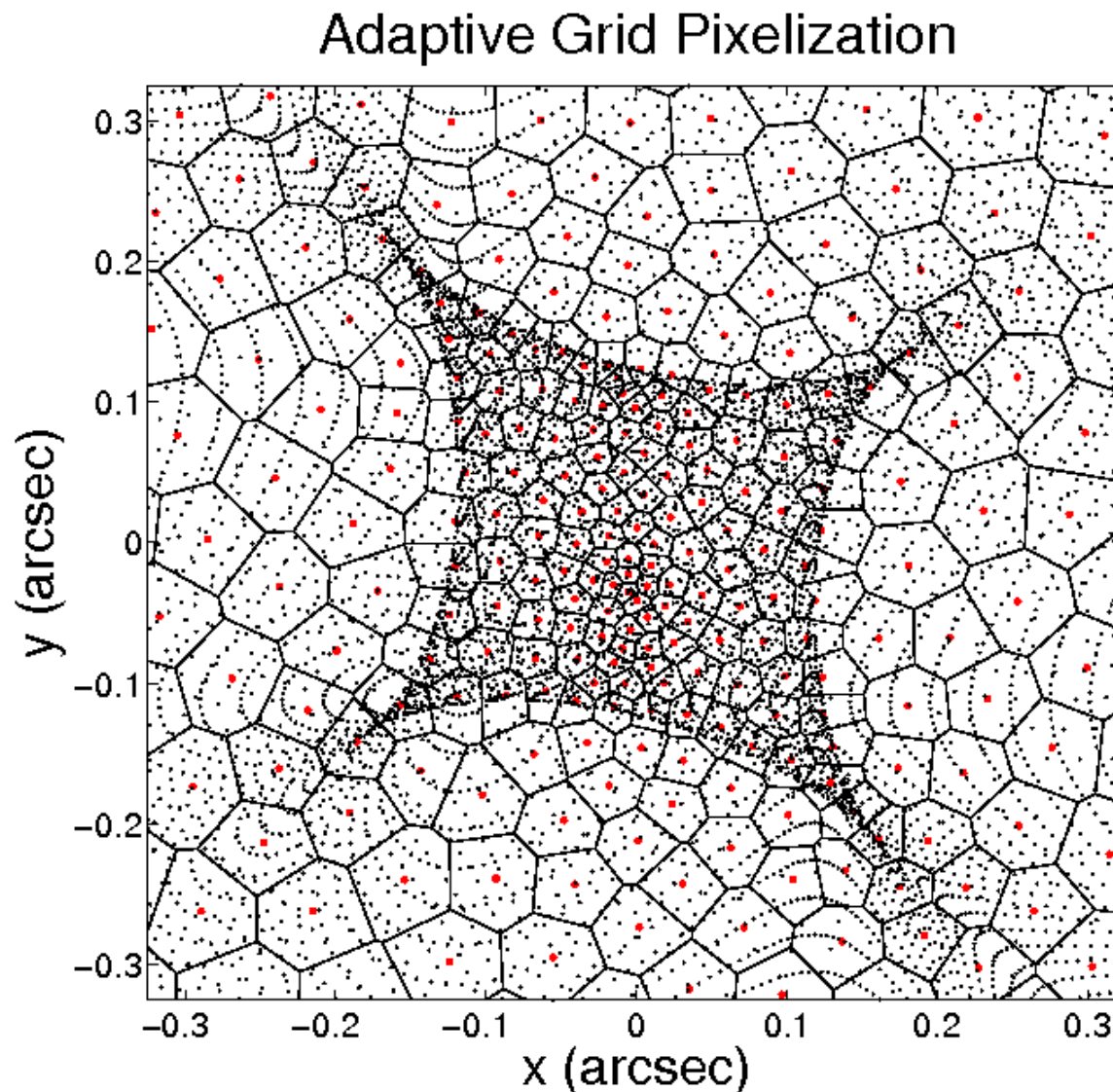
Systematics - Discreteness Bias

- A square grid uses one fixed source plane discretization.
- Systematics associated with this bias the lens model.
- **A problem for all other methods in the literature!**



Systematics - Discreteness Bias

- Evidence In
- 1: -4071.70

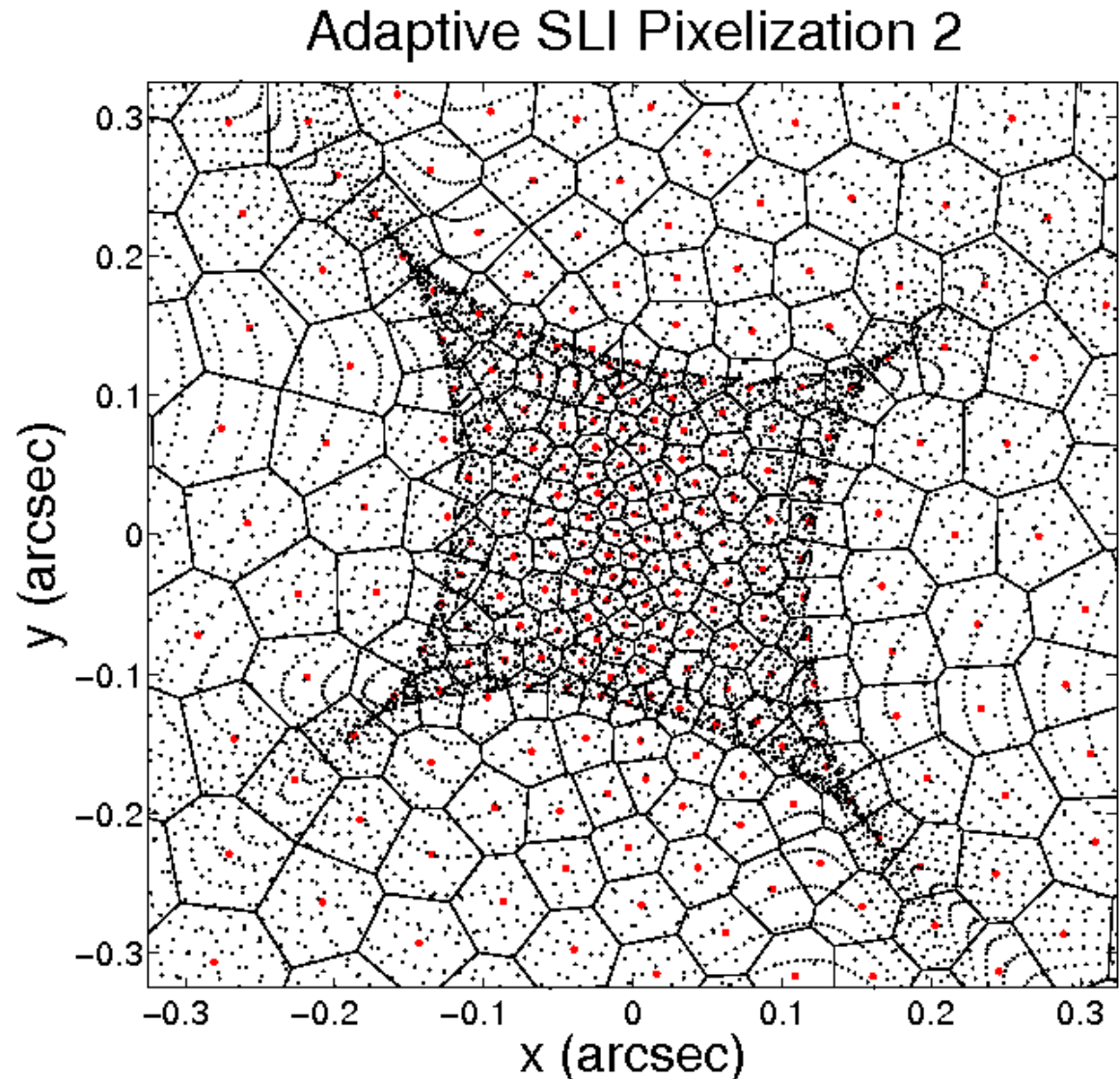


Systematics - Discreteness Bias

- Evidence In

1: -4071.70

2: -4080.31



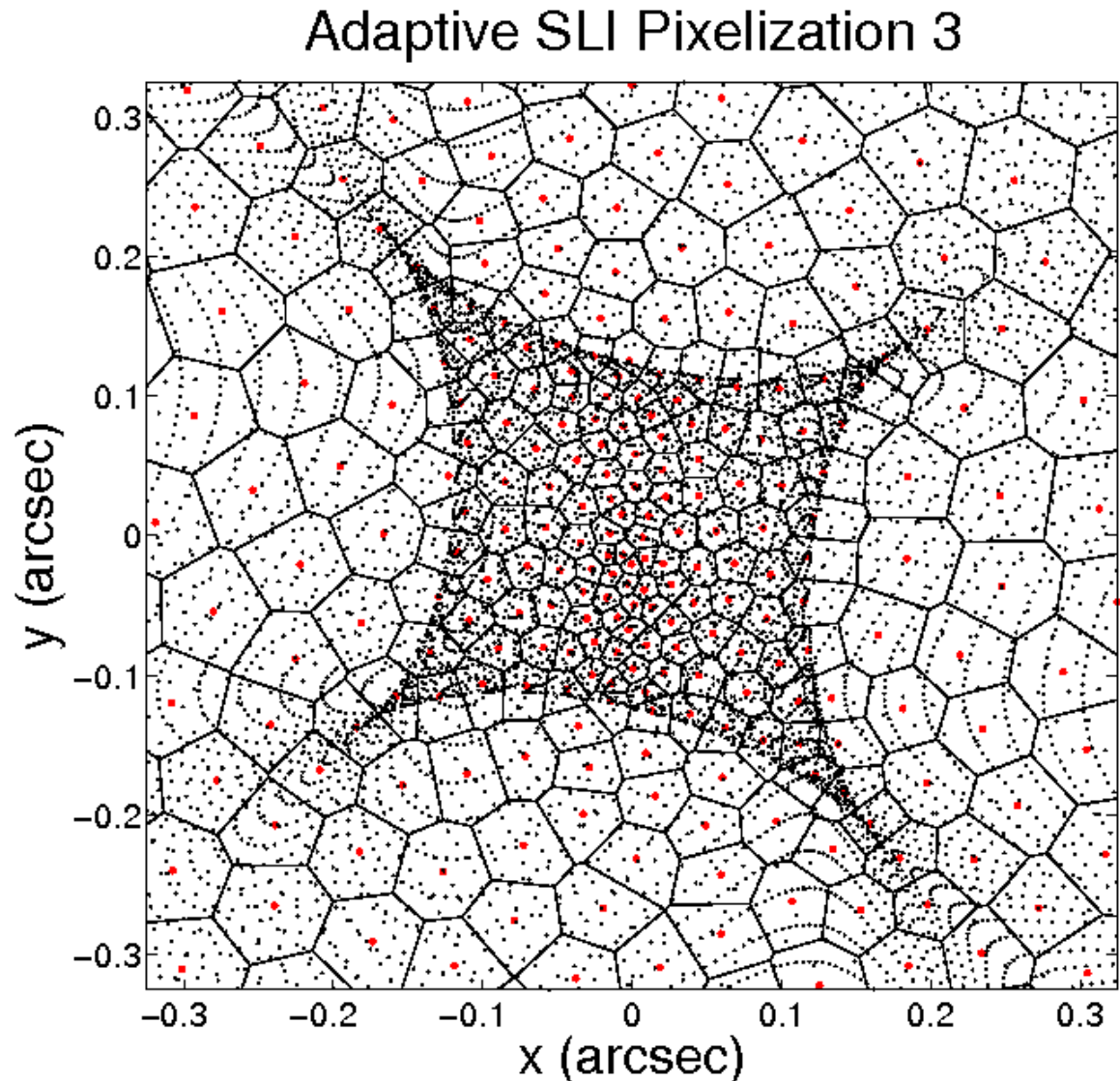
Systematics - Discreteness Bias

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2: -4080.31

3: -4072.72



Systematics - Discreteness Bias

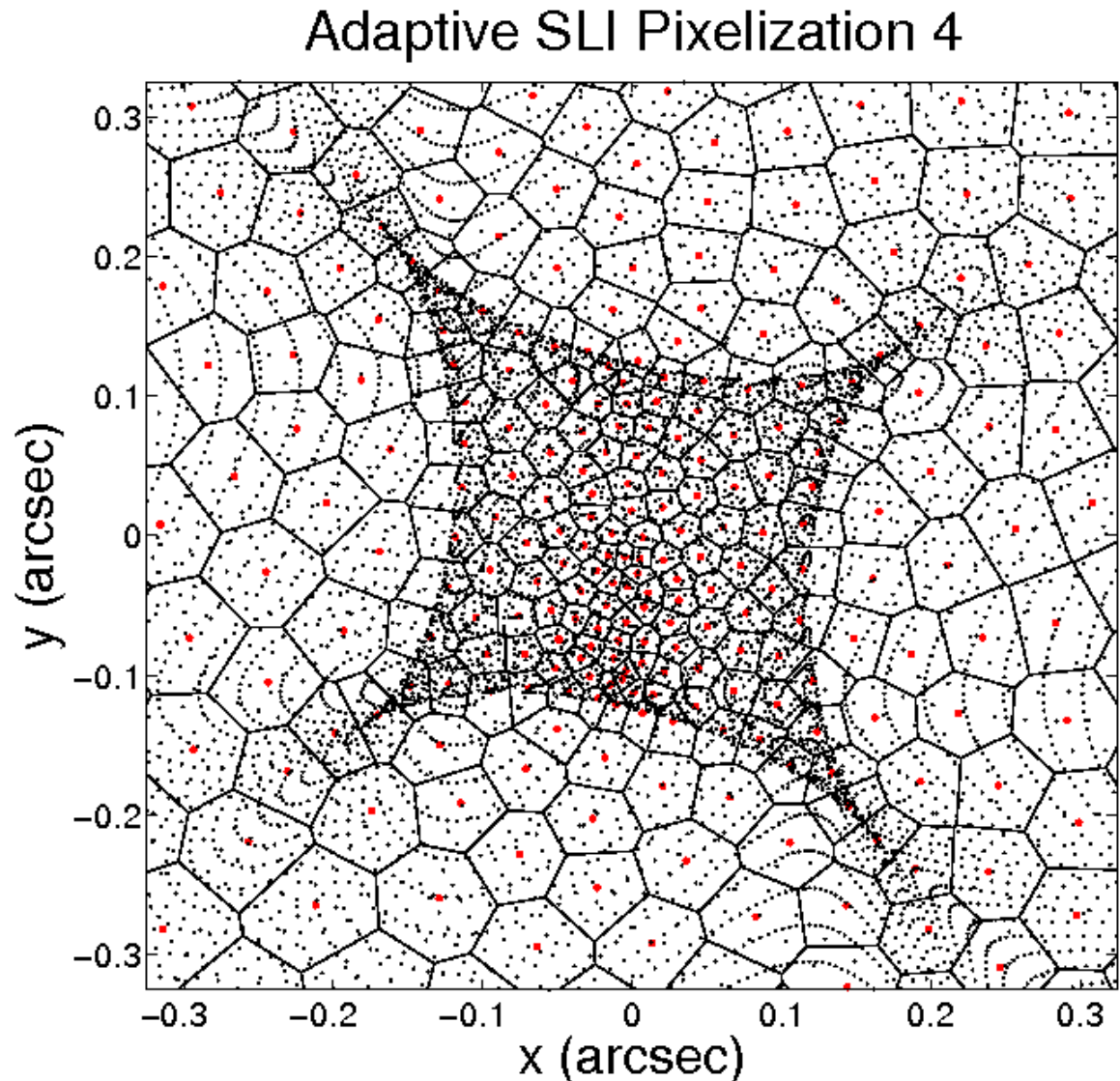
- Evidence In

1: -4071.70

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4: -4067.92



Systematics - Discreteness Bias

- Evidence In

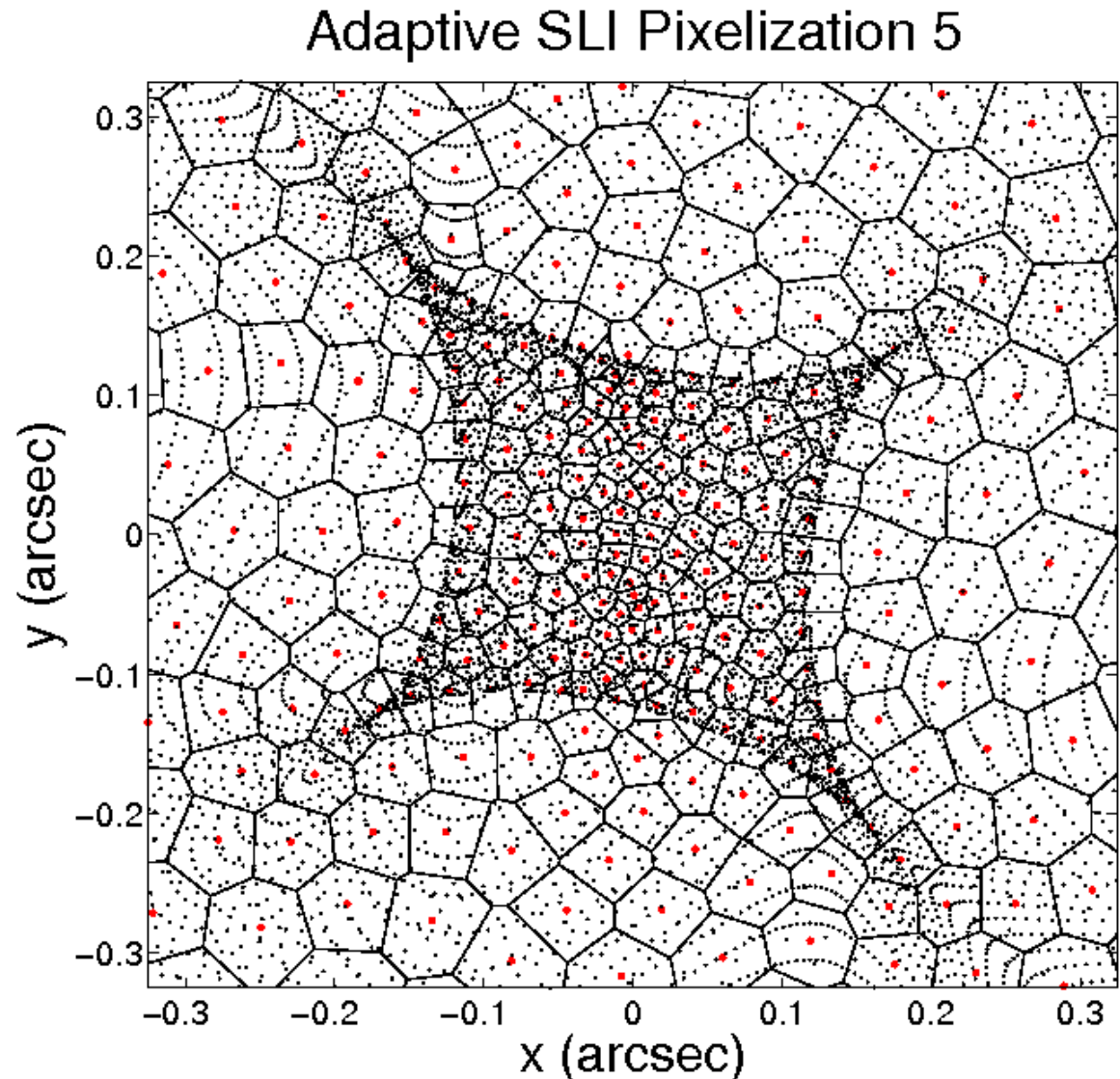
1: -4071.70

2: -4080.31

3: -4072.72

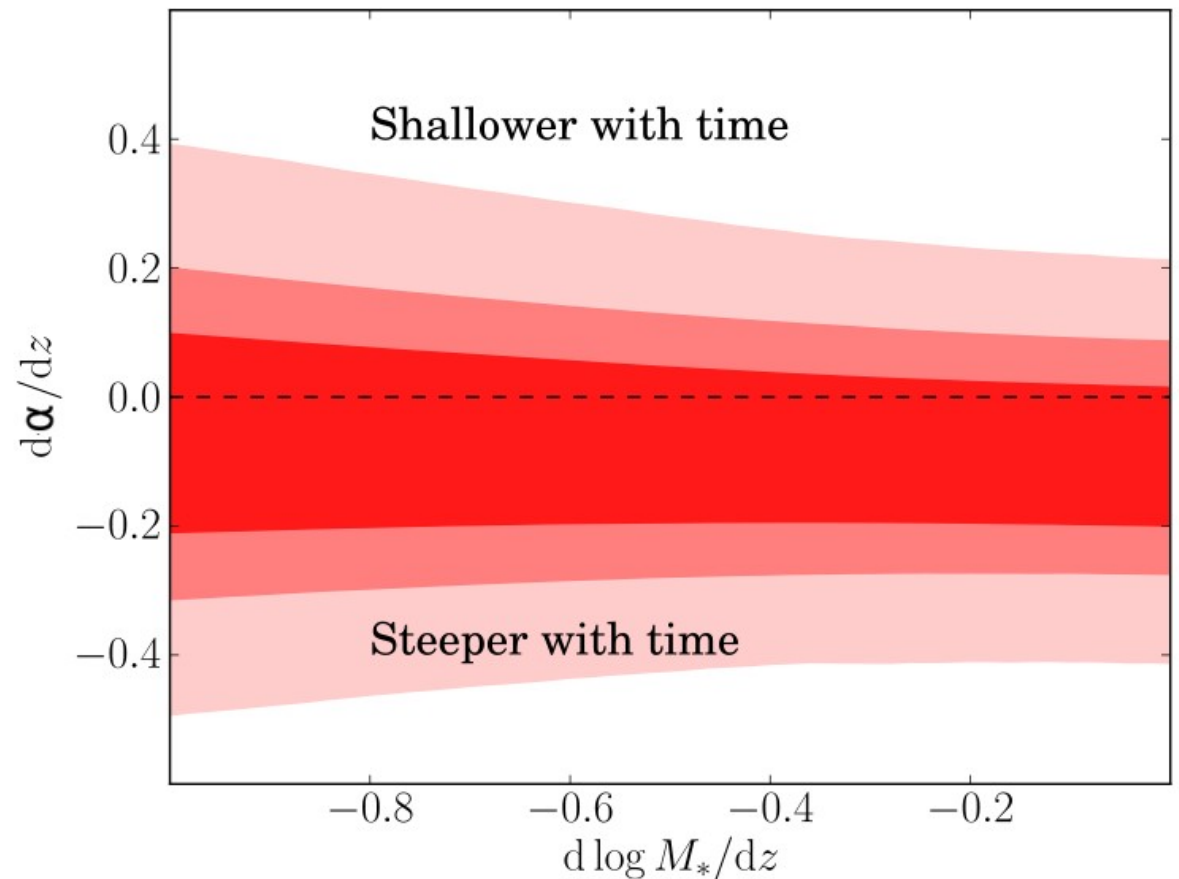
4: -4067.92

5: -4056.31



Lensing + Stellar Dynamics

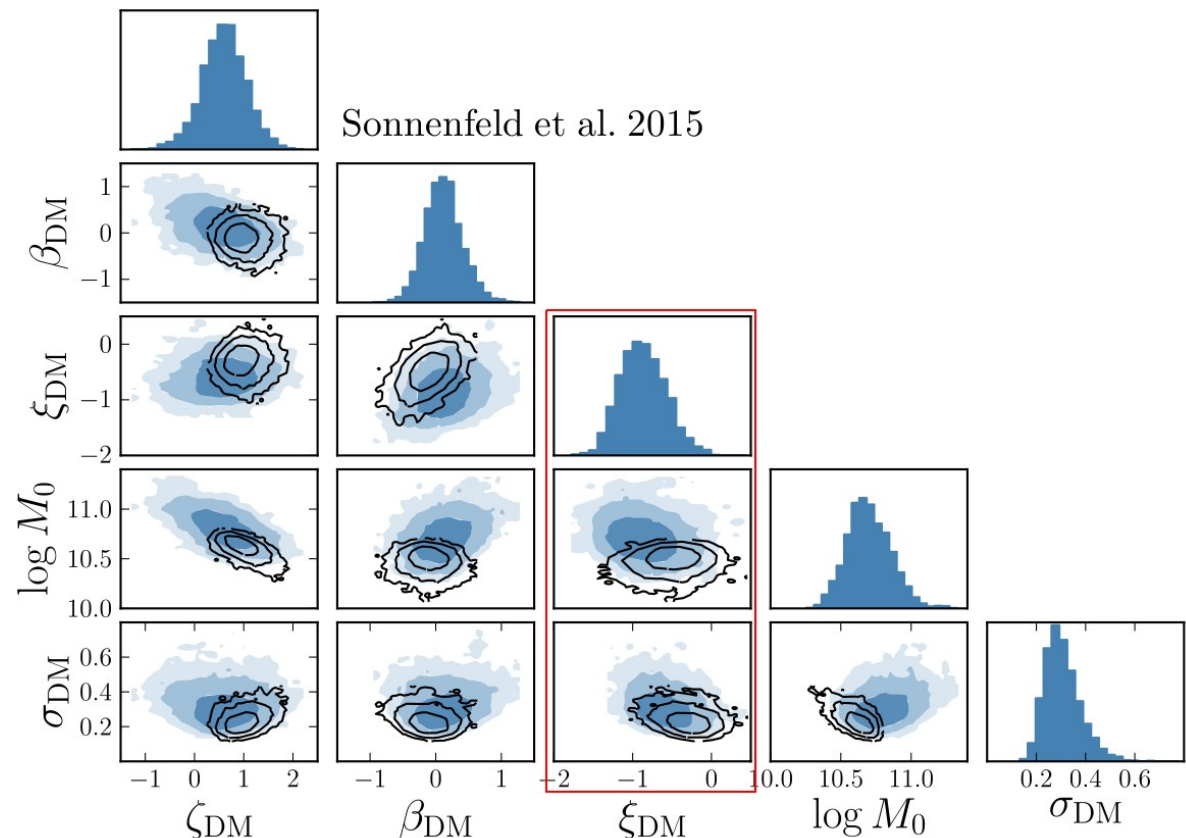
- Account for change in stellar mass density using mass-size relation.
- No evolution in an ETG's value of α .



Sonnenfeld et al. 2014

Lensing + Stellar Dynamics

- Decomposed (light + dark) mass models.
- 3 parameters, 2 observables.
- Anti-correlation between stellar mass density and halo mass.



Why so slow?

- Lens Modeling demands non-linear analysis throughout.
- Highly complex parameter space, needs careful initialization
- Non-linear sampling performed with **MultiNest**.

Automated Analysis

- ***AutoLens* comprises an (up to) 8 phase analysis pipeline.**
- **Settings changed throughout pipeline – use faster settings early on when precision not required.**
 - > Lower source plane resolution.
 - > PSF-trimming.
 - > Faster non-linear sampling.
- **Tune settings to 'scale up' to large samples.**

Automated Analysis

- **Early Phases – Setup Analysis:**
 - > Setup the analysis (Bayesian Framework, Lens Model, Source Plane)
- **Middle Phases – 'Build' the lens model:**
 - > Choose the light, mass and light / dark matter geometry using model comparison.
- **Later Phases - Precise Lens Modeling.**
 - > Simultaneously model all lens parameters with high precision (slower) settings.